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User Handbook
of
INTELLIGENT VEHICLE INTERCOM
SYSTEM - iVIS
(UCZN10331)

(DS Cat No.) _____

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RECORD OF CHANGES

Change Notice No.	Authority	Incorporated by “Name and Rank” in Block letters	Initials with date
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ASSOCIATED DOCUMENTS

Technical Literature for INTELLIGENT VEHICLE INTERCOM SYSTEM-iVIS (UCZN10331) comprises of the following:

- **USER HANDBOOK**
- **TECHNICAL MANUAL**
 - **PART I: TECHNICAL INFORMATION**
 - Volume 1: Technical Description
 - Volume 2: Drawings
 - **PART II: Maintenance**
 - **PART III: Overhaul and Reconditioning Instructions**
 - **PART IV: Manufacturer's Parts List**
 - Volume 1: Parts List
 - Volume 2: Illustrations

SAFETY WARNING

The voltages employed in this Equipment are sufficiently high to endanger human life.

Power must be switched off before servicing the equipment & great care taken when making internal adjustments etc.

Ringer output Voltage in this equipment is sufficiently high to endanger human life. Power of external battery connections / internal batteries is to be removed before servicing the equipment.

Do not touch the metal parts of the line while the System Switch is in Magneto position and Magneto Call button is pressed.

FOR FIRST AID IN CASE OF ELECTRIC SHOCK SEE OVERLEAF

FIG. 1.



FIG. 2.



FIG. 3.



FIG. 4.



1. SWITCH OFF. If this is not possible, PROTECT YOURSELF with dry insulating material and pull the victim clear of the conductor. DO NOT TOUCH THE VICTIM WITH YOUR BARE HANDS until he is clear of the conductor, but DO NOT WASTE TIME.

2(a) Place the victim in the supine position.

2 (b) Keep the air passages clear by turning the head to one side, opening the patient's mouth and clearing it of water, saline, mucus or blood, a lot of which might have accumulated in the back of the throat.

(Figure 1)

3. 4. If the jaw is rigid try to force the mouth open by pressure on the gum behind the last molar tooth of the lower jaw. When the upper air passages are thus cleared, tilt the head backward and force the jaw forwards from the angles of the jaw in front of the ears. This would prevent mechanical obstructions to the upper air passages.

(Figure 2-3)

(a) Then hold the chin up and forward with one hand and pinch the nostrils of the victim with the other.

(Figure 4)

(b) Take a very deep breath and apply your mouth to that of the victim and blow into his mouth, until the chest of the victim moves up indicating filling of the lungs. (NEVER ALLOW THE CHIN TO SAG).

(Figure 4)

5. When the chest has moved up, withdraw your mouth and allow the chest to sink back. REPEAT this process every three to four seconds until the victim begins to breathe again or until he is taken over by a medical attendant. This method can be continued in an ambulance during transit of the patient from the site of accident to the nearest medical centre. HAVE SOMEONE ELSE AND SEND FOR A DOCTOR. KEEP PATIENT WARM AND LOOSEN HIS CLOTHING. DO NOT GIVE LIQUIDS UNTIL THE PATIENT IS CONSCIOUS.

IMPORTANT SAFETY INSTRUCTIONS

The equipment described in this manual is intended for use only by authorized personnel who are sufficiently trained in the installation and / or operation of electrical equipment.

Please read carefully all of the instructions listed and keep the manual for later reference.



- a) This equipment is designed to operate from a 220 Volt AC power supply. Never connect the equipment to any other voltages.
- b) Do not allow anything to rest or trap the power cord. Do not locate the equipment where persons could walk on the power cord.
- c) Remove the power to the equipment before cleaning. Do not use liquid cleaners or aerosol cleaners. Use a damp cloth for cleaning.
- d) Never open the equipment and change cards or modules with the power connected.
- e) Always use proper ESD protection when performing maintenance inside the equipment.

DEMOLITION OF MATERIAL TO PREVENT ENEMY USE**(A) General**

The demolition procedure outlined below will be used to prevent the enemy from using or salvaging this equipment. **DEMOLITION OF THE EQUIPMENT WILL BE DONE ONLY ON ORDER OF THE COMMANDER.**

(B) Methods of Destruction

Use any or all the following methods to destroy the equipment.

- Smash - Smash the crystals, controls semiconductors, coil, switches, transformers and handset. Use sledges, hand axes, pickaxes, hammers crowbars and heavy tools.
- Cut - Cut the cords, headsets and wiring. Use hands axes or hatchets.
- Burn - Burn the cords, resistors, capacitors, coils, wiring, technical manuals. Use gasoline, kerosene, oil, flamethrowers or incendiary.
- Bend - Bend the panels, cabinet and chassis.
- Explosives - If explosives are necessary, use firearms, grenades or TNT.
- Disposal - Bury or scatter the destroyed parts in slit trenches, fox holes or throw them into streams.

CHANGE NOTICE

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Personal Notes

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1 Leading Particulars and General Data

1.1 General

This User manual is intended for the users of the Intelligent Vehicle Intercom System iVIS, (here after referred to as iVIS). It illustrates 'How to use the iVIS system' efficiently.

This manual is divided into several chapters, each covering different aspects of the iVIS Vehicle Intercom System.

Chapter 1 : This introductory chapter about document.

Chapter 2 : Contains brief technical data of iVIS.

Chapter 3 : Overview of iVIS Vehicle Intercom System.

Chapter 4 : Describes the un-packing and removal of preservatives.

Chapter 5 : Describes the installation and preliminary checks

Chapter 6 : Describes the operating instructions.

Chapter 7 : Describes the configuration of iVIS

Chapter 8 : Provides document history and change note.

1.2 Abbreviations

BITE	Built-in Test Equipment
CMDR	Commander
CNR	Combat Net Radio
DC	Direct Current
DIR	Direct Intercom Mode
EMC	Electromagnetic Compatible
EMG	Emergency Call Mode
EMI	Electromagnetic Interference
ESD	Electrostatic Discharge
GNSS	Global Navigation Satellite System
IC	Intercom Mode
IP	Internet Protocol
KBRD	Keyboard
MGMT	Management
MTBF	Mean Time Between Failure.

MTTR	Mean Time to Repair.
Msg	Message
Msgs	Messages
OLED	Organic light-emitting diode
PC	Personnel Computer
PTT	Press to Talk
RAD	Radio Mode
RH	Relative Humidity
SPK	Speaker Unit
VOL	Volume

1.3 Disclaimer

Various figures shown in this document are for indicative purpose. Actual product might be little different from the figures.

2 Intelligent Vehicle Intercom System (iVIS)

iVIS is an Intelligent Vehicle Intercom System, customized for Tracked as well as Wheeled Platform based Armoured vehicle deployment.

iVIS provides facility for commander and various other crew members to communicate with each other. It also provides facility to commander and users to communicate through radio sets interfaced with iVIS. iVIS has a facility of Stick Radio communication to allow any crew to dismount and stay connected with the in-vehicle crew.

With the provision of speaker connectivity and alarm interface, Intercom System provides the complete solution for vehicle intercom system.

iVIS has provision of GNSS location display through ALNS connectivity.

iVIS systems support easy to operate blind operation by way of use of mechanical rotary switch for direct selection of various modes like Intercom /Radios and headset volume controls. System support audio indication of rotary switch position change for full featured blind operation.

2.1 Salient Features

- a) Compact, lightweight, and ruggedized.
- b) Easy to install, operate and maintain.
- c) Operates from vehicle power/external battery of 24V DC.
- d) All the User Units are identical hardware & software wise.
- e) For the ease of understanding, the name of the user unit can be configured as per the user profile. For example; name of the user unit, used by commander, can be configured as Commander Unit. User Unit, used by Second Commander Unit, can be configured as Sec Commander Unit & so on.
- f) User friendly headsets with Active Noise Cancellation (ANR)
- g) Push to Talk headset function for speech transmit.
- h) Provision of auto-PTT (VOX based) operation in Intercom mode.
- i) Available in colours Olive Green

2.2 Technical Specifications

Power Specification	
Input Voltage	▪ Vehicle power operable ▪ 12-36 Volt DC
Power Consumption	CSU : 15W UU : 6W
Power Protections	▪ Reverse voltage protection ▪ Short circuit protection ▪ Open circuit protection ▪ Surge voltage protection ▪ Lightning protection ▪ Basic EMP protection
Standard	Standard:MIL-STD-1275E
Dimension <i>(For details, Refer section 3.2.1& 3.3.1)</i>	
Dimensions (LxWxH)	CSU : 210x197x86 mm UU : 157x180x84 mm (Max)
Mounting	Vehicle Mountable
Interfaces	
Radio Ports	▪ Max of 4 radio interfaces with CSU ▪ Optional RS232 interface to Radios for command and control
STICK Radio	STICK Radio with CSU Range upto 1500 meters
Ethernet Ports	▪ 5 No. eth interface available with CSU ▪ 1 No. eth interface available with every UU
Single Pair Ethernet (SPE)	▪ 1 No. SPE interface available with CSU ▪ 1 No. SPE interface available with every UU
Headset Interface	▪ 1 Headset port with every User Unit
Utility Ports <i>(Optional)</i>	▪ RS232/RS485 and CAN ▪ Alarm Sensors (max 8) ▪ Magneto Field phone connectivity
User Controls	
Display	20x4 OLED display
Volume Control	Rotary volume control to adjust Recv level of Headset speaker
Mode Selection	Mode Selection knob for Intercom Call or to select the radio set.

Accessories	
HEADSETs	▪ Ruggedized audio interfaces featuring advanced noise cancellation ▪ ANR/PNR variants with bail out ▪ Position PTT (On/Off/Momentary) ▪ Operating in high noise environment ▪ ANC on/off and Talk through facility
QWERTY Keyboard	▪ QWERTY Keyboard connects to any User Unit ▪ Used to send text messages and perform basic configuration and maintenance. ▪ Features adjustable, auto-off backlighting. ▪ Multiplexed functions keys with number row. ▪ Function keys to perform special functions
Maintenance	
Power On Self-Test	Performs automatically & shown alarms on display of every UU
Self-Diagnostic	Built In Test Equipment (BITE)
Environmental Characteristic	
Operating Temp	-20°C to 55°C
Storage Temp	-40°C to 70°C
Relative Humidity	0 – 95 % @ 40°C
High altitude	Up to 5000 mtr above sea level

Table 1: Technical Specification of iVIS

3 Brief Technical Description

3.1 About iVIS System

iVIS is the Vehicle Intercom System, customized for Tracked as well as Wheeled Platform based Armoured vehicle deployment. iVIS provides facility for its users to communicate with each other. It also provides facility to its users to talk through in-service CNR radios also.

iVIS intercom system consists of following sub-components:

- [1] Central Switch Unit (CSU)
- [2] User Units (UU)
- [3] QWERTY Keypad Unit
- [4] STICK Radio & Base Station Antenna
- [5] User Headsets
- [6] Interconnection Cables

iVIS is an IP based system wherein Central Switch Unit (CSU) acts as heart of the system & it controls all the interfaced user units. User Units communicate with each other through CSU via ethernet interface. CSU & all User Units have power and ethernet interface with them.

CSU & User Units have one more special ethernet interface i.e. SPE (Single Pair Ethernet) which is used to extend the data connectivity over reduced wiring for use through SLIP RING of vehicle.

3.2 Central Switch Unit (CSU)

The iVIS Central Switch Unit (CSU) is a centralized control unit which controls User Units, Radio Sets & STICK Unit. All the units remain connected to the CSU through interconnecting cables.

CSU is operated from the vehicle power & for rest of the user units, CSU provides desired power & data through interconnecting cables. Desired configuration of the iVIS resides in CSU & user units also.

Central Switch Unit of iVIS component is responsible for distributing power and establishing the IP network for interconnection of all other components of Intercom System.

Central Switch Unit has a provision to interface 4 In-service Radio Sets. Additionally, CSU has provision of one radio base station (along with

Antenna). If equipped Users can also communicate with dismounted soldier/Stick Radio wirelessly.

Radio Sets & STICK Radio can be accessed by CMDR or other Users. Radio & STICK Radio access is configurable.

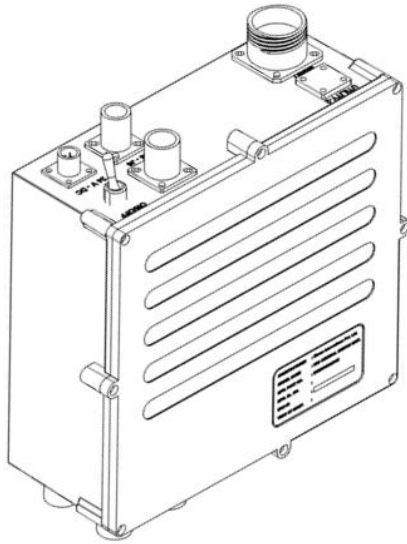


Figure 1: CSU - Wall Mountable

CSU is a wall mountable unit which will be mounted on a plane surface (using a mounting plate) in vertical direction as shown in Figure 1.

Figure 2 shows front view of CSU when CSU is kept on a tabletop. Figure 2 shows rear view of CSU when CSU is kept on a tabletop.

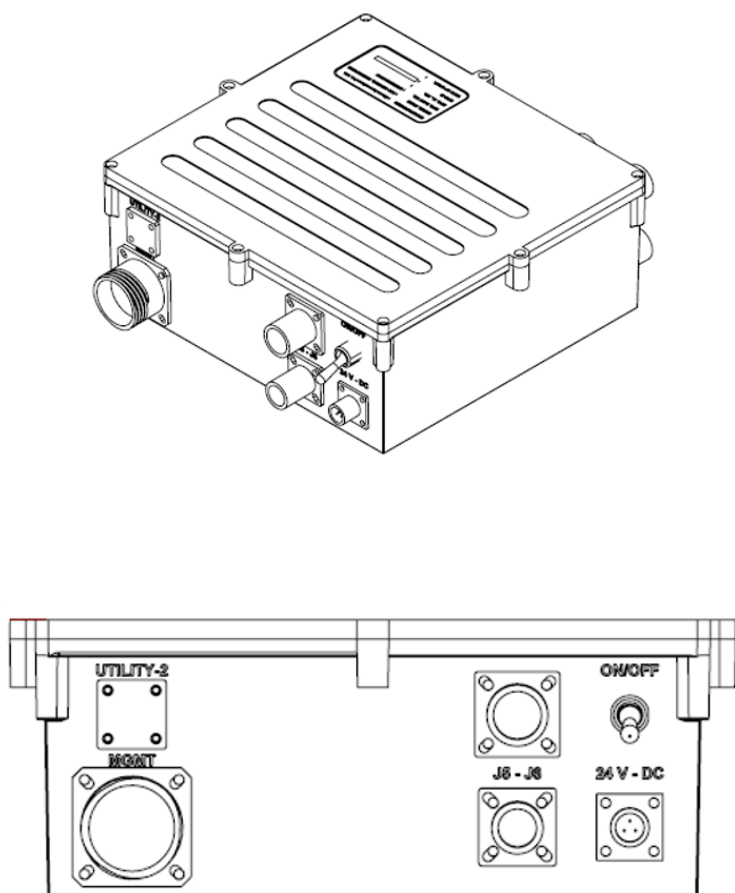


Figure 2: CSU – On a Tabletop (Rear Connectors)

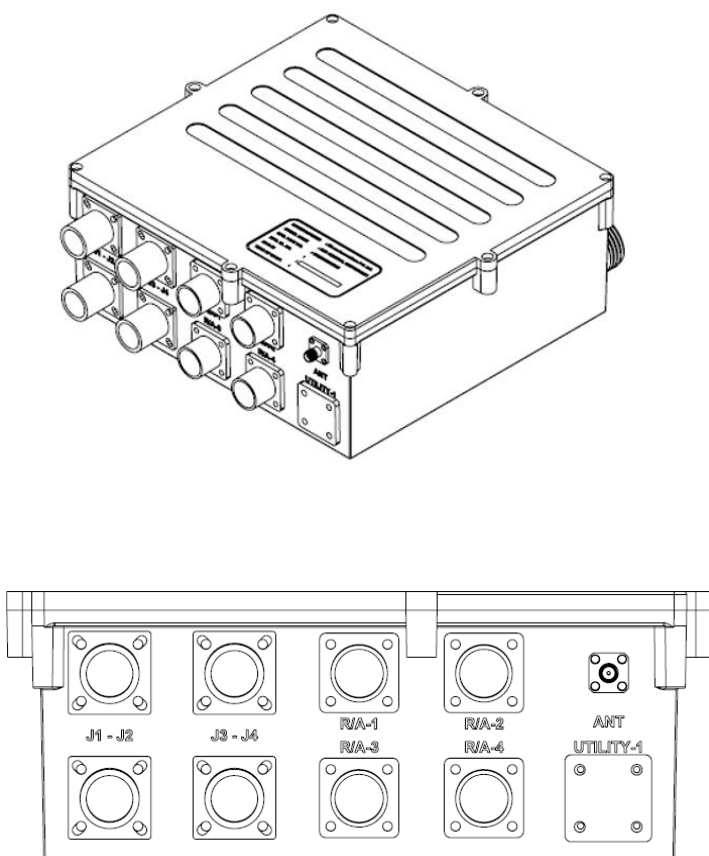


Figure 3: CSU – Isometric View on Tabletop (Front Connectors)

3.2.1 CSU Dimensions

Table 2 shows all the dimensions of CSU. For the ease of understanding, dimensions are taken considering unit is wall mounted & all connectors are Up or down facing as shown in Figure 4.

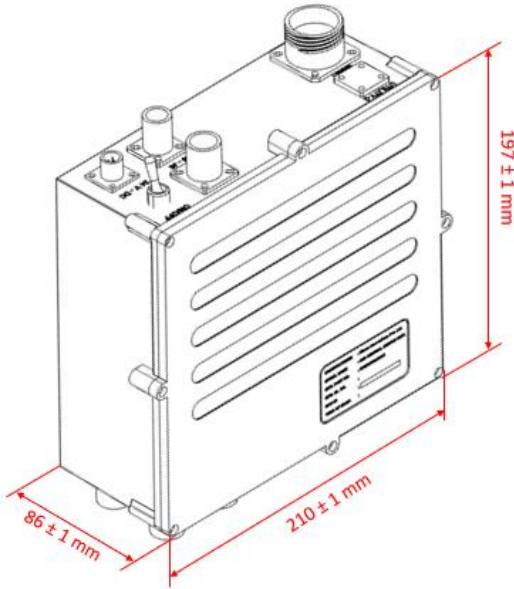


Figure 4: CSU Dimensions

Dimension	Measurement
Length	210 mm (± 1 mm)
Height	197 mm (± 1 mm)
Depth	86 mm (± 1 mm)
Weight	2.25 Kg (± 200 grams)

Table 2: CSU Dimension

3.2.2 Interface Connectors on CSU

Table 3 shows all the interfaces available with CSU;

Interface Connector	Use	Location
24V-DC	Input Power Connector	Top Panel
ON/OFF	Power ON/OFF Switch	Top Panel
J5	5 th Data Power Connector	Top Panel
J6	Single Pair Ethernet Connector	Top Panel
MGMT	Management Interface Connector	Top Panel
UTILITY-2	2 nd Utility Port	Top Panel
J1 to J4	1 st , 2 nd , 3 rd & 4 th Data Power Connector	Bottom Panel
R/A-1 to R/A-4	Radio Interface Connectors - 4 No.	Bottom Panel
UTILITY-1	1 st Utility Port	Bottom Panel
ANT	STICK Radio ANTENNA Connector	Bottom Panel

Table 3: Interface Connectors on CSU

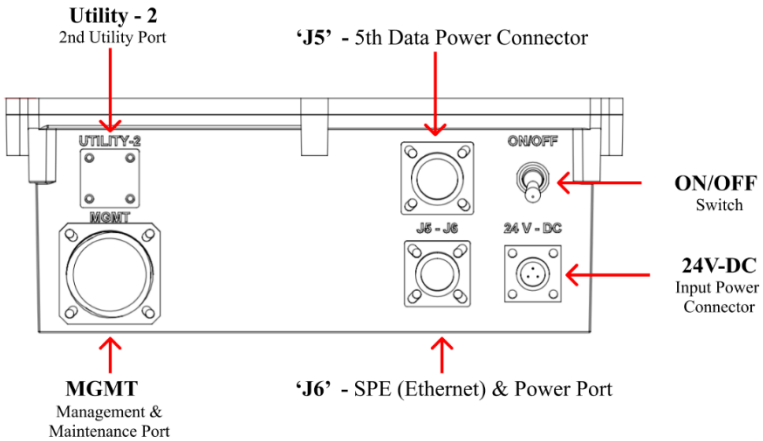


Figure 5: Interface Connectors on the Rear of CSU

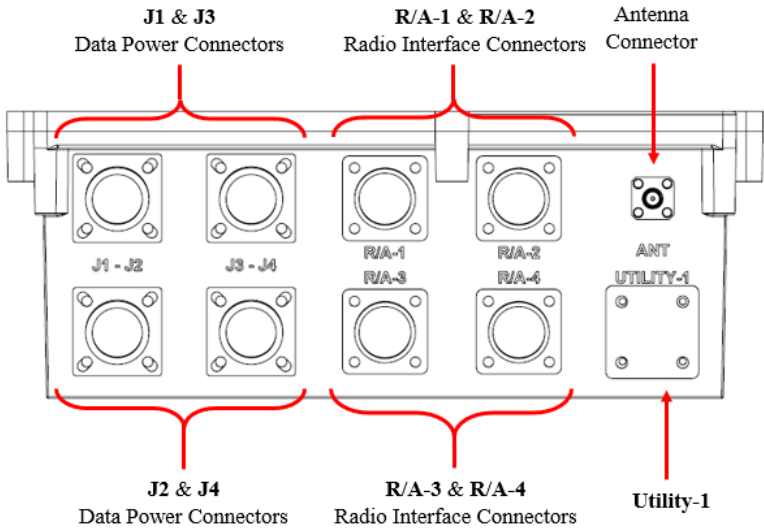


Figure 6: Interface Connectors on the Front of CSU

Figure 5 & Figure 6 shows various connectors available on the top & bottom of CSU. Illustration of all these interface connectors is as follows;

3.2.2.1 Input Power Connector - 24V-DC

The connector 24V-DC is used to feed the input power 24Volt DC to CSU from vehicle battery power. Care to be taken to connect +ve & -ve wires accordingly while connecting 24Vdc from vehicle.

3.2.2.2 Power ON/OFF Switch - On/Off

ON/OFF Switch is to switch off the input power to CSU. Keep the On/Off switch in OFF position while connecting the cable to the 24V-DC connector.

3.2.2.3 Radio Interface Connector - R/A-1 to R/A-4

iVIS has a provision to connect multiple radios. A max of 4 Radio sets can be connected using CSU. Connector R/A-1 to R/A-4 are used to connect four in service Radio Sets with appropriate Radio cables. Radio sets shall be provided by customer.

3.2.2.4 Data Power Connectors for UUs - J1 to J5

Connectors J1 to J5 are the interfaces which are used to connect User Units. These interfaces are used to provide ethernet data and separate power to connected user unit.

3.2.2.5 SPE Connector

This is the special ethernet interface (Single Pair Ethernet) which is used to connect UU to CSU if one unit is in Turret & another is in Hull of the tank. SPE connectivity will be used to extend the ethernet interface and power across Turret.

If SPE interface must be used, a special cable 'SPE Cable' shall be used, to provide ethernet & power.

SPE can also be used to daisy chain of UU's to expand the capacity of iVIS system

Figure 7 shows how the SPE interface is extended through SLIP Ring.

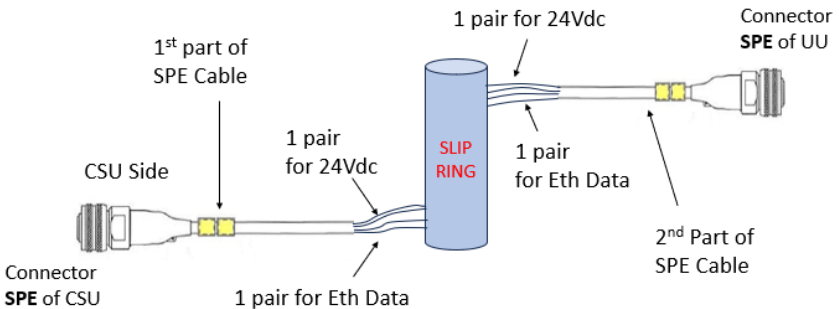


Figure 7: SPE Interface – Eth Extension

3.2.2.6 STICK Radio Antenna Connector – ANT (Optional)

CSU has option of one no. radio base station Radio Set. Connector ANT is used to connect the Antenna. If equipped, an Antenna cable also is provided with iVIS.

3.2.2.7 Management Port MGMT

For configuration, diagnostics, and maintenance purpose, CSU provides a dedicated RJ45 Management **MGMT** (Ethernet connector). A laptop or PC can be connected to this interface to access system management tools, perform configuration updates, monitor system health, and carry out troubleshooting tasks.

3.2.2.8 UTILITY-1 & UTILITY-2, Utility Ports (Optional)

The iVIS system provides 2 No. Utility Connectors on CSU. Every connector has multiple interfaces for integration with external devices or vehicle systems. CSU supports various interface protocols to connect external peripherals such as the ALNS/GNSS system, Voice Warning System or external Speaker tc. Through these utility ports, vehicle-mounted electronic sub-systems are connected.

The Utility Port also offers an audio output to connect an external loudspeaker.

In addition, CSU may provide dedicated digital input interfaces for connecting external digital alarm sources. CSU also supports Magneto Field Telephone connectivity to connect legacy magneto field phone.

3.3 User Units (UU)

The iVIS User Unit (UU) serves as the primary crew interface within the iVIS system. It enables headset-based communication, user interaction, and real-time access to system and operational information. The UU is designed to provide intuitive control, enhanced situational awareness, and reliable communication to the personnel operating in-service vehicle.

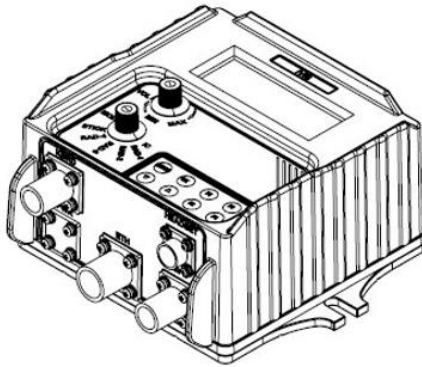


Figure 8: User Units – Isometric View

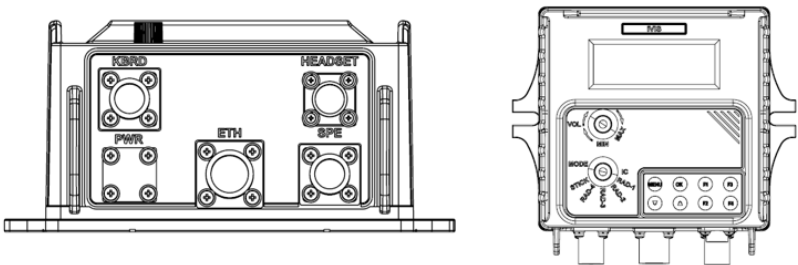


Figure 9: User Units - Top & Front View

iVIS can have multiple User Units & all User Units are identical (hardware & software wise also). Any of the User Unit can be configured for access rights as per the profile of the user. For example, for an armed vehicle, User Units can be configured as;

- | | |
|--------------------------|-----------------|
| ▪ Commander Unit | - CMDR Unit |
| ▪ Section Commander Unit | - SEC CMDR Unit |
| ▪ Section 2IC Unit | - SEC 2IC Unit |
| ▪ Gunner Unit | - GUNR Unit |
| ▪ Driver Unit | - DRVR Unit |

Every User Unit has a built-in keypad which is used for various functions like status view, diagnostics, BITE operation and configuration.

Every user can access various In-service radios or STICK Radio depending upon access rights given to that user. Access rights to user units must be defined through configuration

Configuration of iVIS system can be done by any of the user unit. Some of the configurations are password dependent (refer chapter 7).

3.3.1 User Unit Dimensions

Table 4 shows all the dimensions of UU. For the ease of understanding, dimensions are taken considering unit is wall mounted & all connectors are down facing.

Dimension	Measure with Tolerance
Length	147 ± 1 mm
Length with mounting cuts	180 ± 1 mm
Height without protective hinges	142 ± 1 mm
Height with protective hinges	157 ± 1 mm
Depth	84 ± 1 mm
Weight	1.5 Kg ± 100 gm

Table 4: UU Dimension

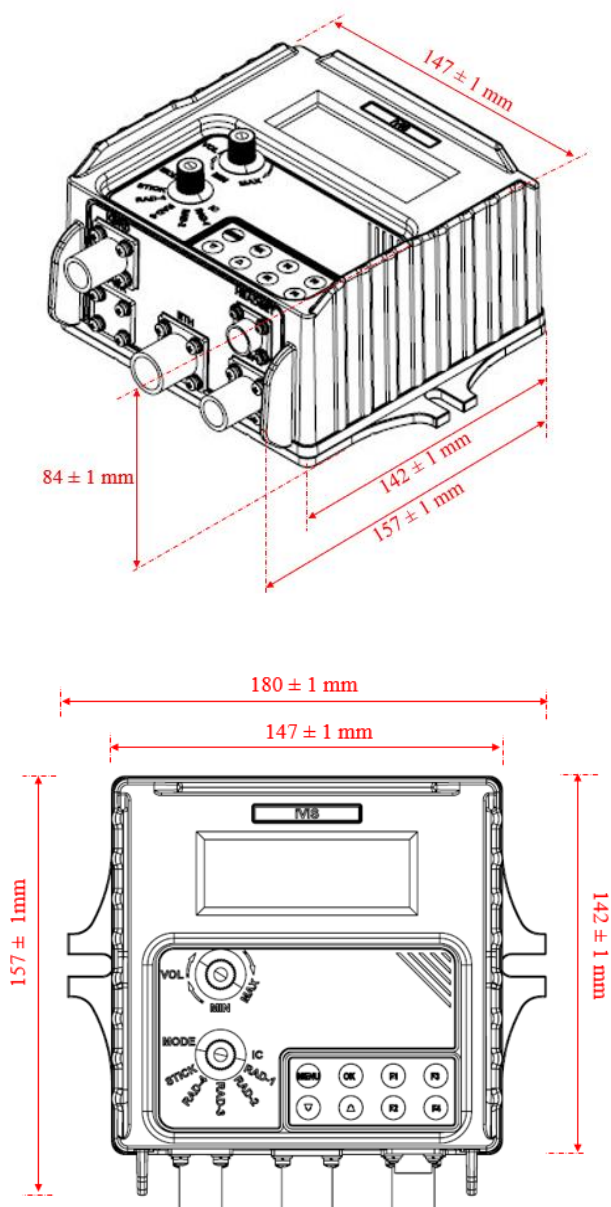


Figure 10: UU Dimensions

3.3.2 Control & Connectors on User Units

Table 5 shows all the control switches, connectors & display available on User Units.

Control / Connector	Use	Location
Display	OLED 20x4 display	Top Panel
VOL	Volume Switch	Top Panel
MODE	Mode Selection Switch	Top Panel
Function Keys	Function Keypad	Top Panel
KBRD	Keyboard Connector	Top Panel
ETH	Data & Power Connector	Front Side Pannel
HEADSET	Headset Connector	Front Side Pannel
SPE	Single Pair Ethernet Connector	Front Side Pannel

Table 5: Control & Connectors on User Unit

Figure 11 illustrates various connectors available on User Unit.

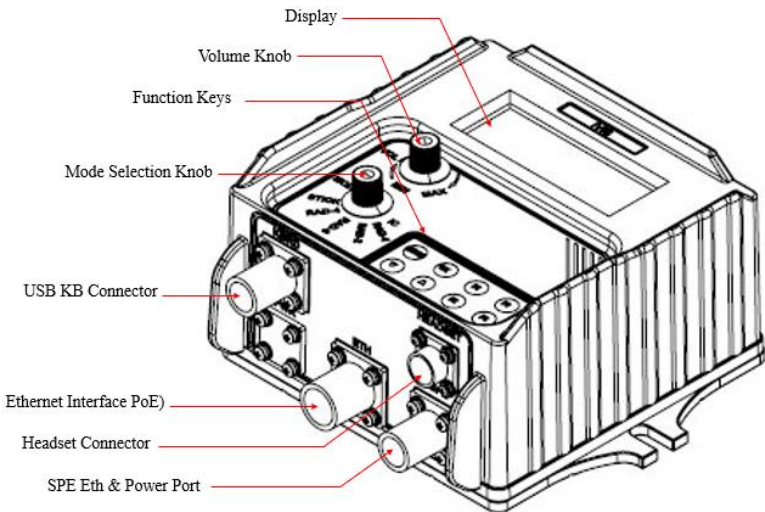


Figure 11: Control Switches, Connectors on User Unit

3.3.2.1 OLED 20x4 Display of User Unit

Every user unit has a 20x4 Alphanumeric OLED Display for clear visibility and user-friendly operation. It is designed to withstand the rigors of road travel, with durability and resistance to harsh environmental conditions.

It facilitates real-time status of user unit showing various information such as the mode selection, active listening connections, volume controls, and health status. It may also provide various alarms and fault indication with unit names, vehicle diagnostic data, and other essential information.

It is designed to withstand the rigors of road travel, with durability and resistance to harsh environmental conditions. User-friendly and robust, the display is an integral part of iVIS system.

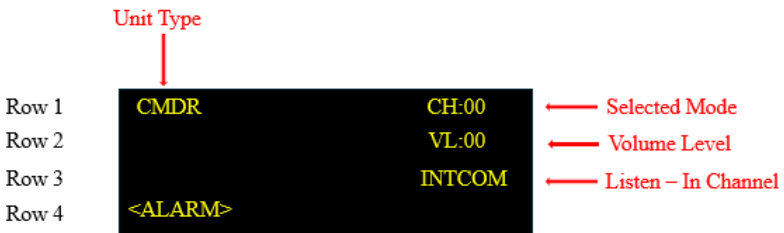


Figure 12: Display Content in Idle Mode

OLED display provides following information:

User Unit Type : Row 1, left side; It will show the name/profile of the user as configured.

Channel No. 'CH' : Row-1, right side, Value 00 to 05
00 for INTCOM
01 for RAD-1
02 for RAD-2
03 for RAD-3
04 for RAD-4
05 for STICK Radio

Volume Level 'VL' : Row-2, right side, It shows 00 to 10 for volume knob position of the Unit. 00 for min, 10 for max.

Receive Channel : Row-3, right side; shows the name of all the channels from where the headset can receive/listen-in the voice in selected operation mode, such as INTCOM, RAD-1, etc. Listening of the voice depends upon the access rights also.

<ALARM> : Row-4, left side; displays in rotation various alarms (if any) as below;

- Unit level Alarms - Status of UUs 'UP' or 'DN'
- Headset Level Alarms - PttStuck/AUD FAULT.
- Vehicle Alarms - If equipped, alarm text, if there is any active alarm with its acknowledgement state as A1: ACTV (ACK or PM-ACK).

If Alarm are configured but alarm hardware is not detected then A1:HW NOT DTD. Normally this field shall be blank if there are no alarms and no faults.

3.3.2.2 VOL Knob

Every User Unit has one ANR Headset associated with it. VOL knob is used to adjust the receive volume level of the speaker volume of the associated headset.

The knob increases the volume when rotated in clockwise direction and decreases the volume when rotated in anticlockwise direction. The knob is positional knob means the knob has fixed number of positions up-to max 10 positions only. The knob is located on the top panel of the unit. An tone is played in the headset speaker for blind operation support on every knob change.

Following picture Figure 13 shows the Headset volume knob.



Figure 13: VOL Knob of User Unit

3.3.2.3 MODE Selection Knob

iVIS has multiple modes of operation as explained in this section. Figure 14 shows the picture of **MODE** selection knob.

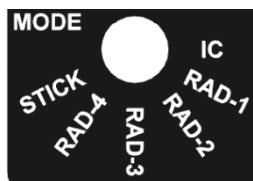


Figure 14: Mode Selection Knob

The description of all the available modes is as follows;

- a) **Intercom** Mode – All users can talk with each other in conference mode. All the interfaced users whose mode is set to **IC** will be able to communicate to each other.
- b) **Radio** Mode – iVIS has a provision to interface a 4 Radios through CSU. User Units, if configured, can communicate to any desired Radio set in half duplex by setting the MODE knob to desired Radio i.e. RAD-1, RAD-2 & soon.
- c) **STICK** Mode – If equipped with Stick Radio Communication feature, User Units whose mode is set to STICK shall be able to have a half-duplex communication with dismounted soldier, if configured. *For more details, refer section 5.3.*

3.3.2.4 Function Keypad

Every User Unit has a fixed **Function Keypad** on the top panel as shown in Figure 11. This keypad is used for status view, diagnostics and configure User Unit.

Function keypad is used in conjunction with the display of User Unit. When User Unit is in Idle state, on pressing **MENU** key, display of User Unit shows certain configuration. **Arrow** keys, **OK** & **F1-F4** keys are used to select & set the desired configuration parameter. Detailed description shall be explained in upcoming section of this handbook.

A close view of the keypad also is shown in Figure 15.

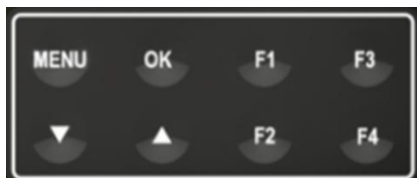


Figure 15: Function Keypad

Various keys and their function are as below:

- | | | |
|----------------------|---|--|
| MENU Key | - | Used to access the menu in 'Idle' mode and back function while in any menu operation. |
| OK Key | - | To select the desired option or action. |
| Down Arrow | - | Go down in selected menu, sub menu or operation. |
| UP Arrow | - | Go up in selected menu, sub menu or operation. |
| F1 to F4 Keys | - | 4 Function Keys, are used to do the iVIS configuration, as navigation keys for text input when in any menu OR as short-cut keys when in IDLE mode. IDLE mode Short cut functions of these keys are as follows

F1 : To Radio Access Control
F2 : To view GNSS details
F3 : To view Radio message options
F4 : To view alarms and acknowledge the alarms |

3.3.2.5 KBRD Connector

iVIS comprises of QWERTY Keyboard which is used to customize SMSs as well as alternate to Keypad present on the User Unit. QWERTY Keyboard is connected to that User Unit. The connector **KBRD** available on User Unit, is used to connect QWERTY Keyboard.

3.3.2.6 ETH Connector

This is the interface connector of User Unit to connect to CSU. Through this interface UU is connected to CSU to get operational input power & data connectivity.

3.3.2.7 SPE Connector

This is the special ethernet interface (Single Pair Ethernet) which is used to connect UU to CSU if one unit is in Turret & another is in Hull of the tank. SPE connectivity will be used to extend the ethernet interface and power across Turret.

3.3.2.8 HEADSET Connector

Every User Unit has one ANR headset connected to it. Connector HEADSET shall be used to connect ANR Headset.

3.4 QWERTY Keyboard

The **QWERTY Keyboard** which is USB keyboard also is the part of iVIS. The keyboard allows users to efficiently compose and edit text messages, significantly improving usability for messaging, data entry, and command execution when compared to keypad-based input.

Though this device can be connected to any of the UU however generally it is used by commander as user input device to enhance text entry and command input capabilities within the iVIS system.

QWERTY Keyboard comes with built-in cable which is directly connected to **KBRD** connector of User Unit.

Refer Chapter 7 for the detailed usage of QWERTY Keyboard Unit.

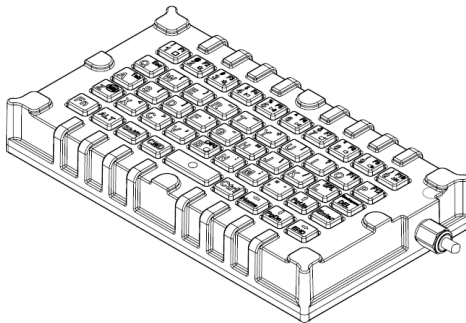


Figure 16: QWERTY Keypad Unit – Isometric View

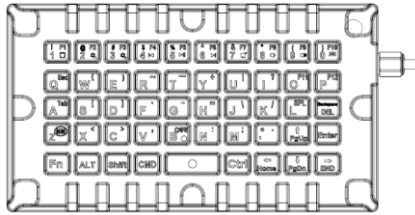


Figure 17: QWERTY Keypad Unit – Top View

3.4.1 Keyboard Dimensions

Table 6 shows all the dimensions of Keyboard Unit.

Dimension	Measure with Tolerance
Length	136 ± 1 mm
Width	76 ± 1 mm
Height	20 ± 1 mm
Weight	0.270 Kg $\pm$ 0.05 Kg.

Table 6: Keyboard Dimension

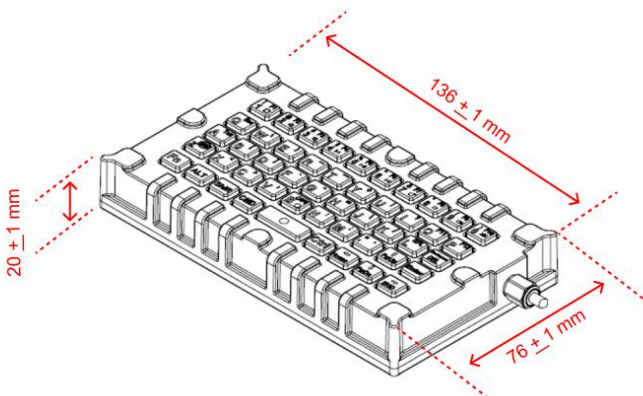


Figure 18: Keyboard Dimensions

3.5 iVIS Headset – ANR Headset

The iVIS User Units support Active Noise Reduction (ANR) / Active Noise Cancellation (ANC) headsets to ensure clear and reliable voice communication in high-noise military vehicle environments.

The ANR headset actively reduces ambient and vehicle noise, improving speech intelligibility and operator comfort during prolonged operation.



Figure 19: iVIS Headsets

PTT Operation on HEADSET

Each supplied Headset has PTT option on left ear-cup. These PTTs are integrated part of headset.

There are two different PTT positions as shown in the Figure 20 below. Following section describes the detailed PTT operation.

a) PTT Momentary

The momentary PTT is used to initiate own voice transmission to the selected user/radio or group of users (selected destination) as per the selected mode of operation. Whenever PTT is pressed own voice will be transmitted to selected destination immediately and others can then listen to the voice. This operation is available in all the modes of operation.

b) PTT Latched or Permanent PTT

Permanent/Latched PTT button is used to stay in intercom mode and talk whenever required without pressing and releasing the PTT. i.e.

this facility enables the crew member hands-free operation to stay and talk in intercom mode.

- a. When the PTT is latched rearward the voice of the crew is transmitted when the crew speaks with enough voice, hence in this mode the crew must speak a little louder to enable his voice transmission.
- b. When the PTT is in this position then crew members mode knob will be rendered in-effective and the crew will remain in intercom mode permanently if the PTT is rearward latched.



Figure 20: PTT positions on left ear-cup

Noise Control and Listening Modes

The ANR headset supports dual listening modes to balance noise reduction and situational awareness that can be switched using mode toggle switch located on the right earcup of headset. Headset modes are as follows:

- **Forward Latch:** Talk through Mode. Allows controlled ambient sound pickup while maintaining communication capability. This mode shall be used only when vehicle is off and user is wearing the headset
- **Centre position:** Off (Both Talk through and ANR off)
- **Backward Latch:** ANR Mode. Actively suppresses ambient and vehicle noise to enhance speech clarity.

3.6 iVIS - STICK Radio & Antenna

iVIS provides an optional STICK Radio, integrated within the CSU, to provide the communication with dismounted soldiers operating outside the vehicle. If equipped this feature allows any in-vehicle user to communicate with Stick Radio using the intercom system.

To provide enough coverage to Stick Radio Communication, a Base Station Antenna is provided as a part of iVIS which remains connected to CSU.

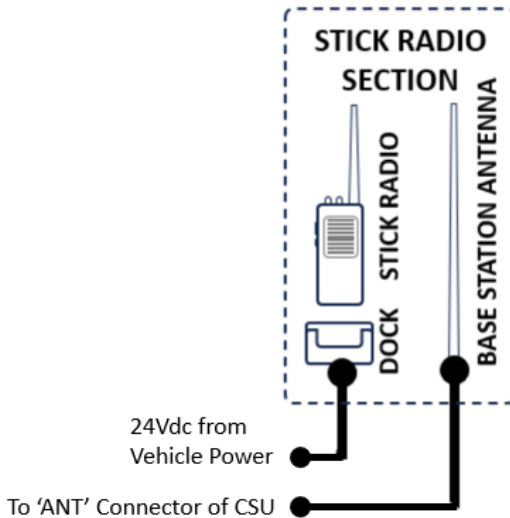


Figure 21: Stick radio & Base Station Antenna

3.7 Interconnection Cables

Every set of iVIS will have following cable set as mentioned in Table 7. Each Cables are identified by cable code and length as follows CBLXXXXX/YYYY where

- XXXXX stands as the cable type as per table below
- YYYY refers to the length of cable in cm.

S/N	Cable	Cable Description	Qty
1	CBL11006 Input Power Cable	Used to feed the Input 24V DC power to CSU from Vehicle Power	01
2	CBL21009 Radio Cable	Used to connect Radio Sets (Radio Model CNR900) to CSU	As required
3	CBL41001 Antenna Cable	To connect the STICK Radio Antenna to the CSU.	As required
4	CBL11006 Doc Station PWR Cable	To feed the input power to docking Station from vehicle power	As required
5	CBL21051 Data PWR Cable	To connect UUs to CSU	As required
6	CBL21052 Headset Cable	Used to connect Headsets to respective User Unit	As required
7	CBL21055 SLIP Ring Cable	Cable if SPE (Single Pair Ethernet) port is used to extend ethernet interface through SLIP Ring	As required

Table 7: iVIS Interconnecting Cables

4 Unpacking & Removal of Preservations

4.1 Packaging Concept of iVIS

All the iVIS components and cables are transported in a packing box. The equipment is ruggedized military equipment and adequate protection for displays, knobs and various connectors is already considered during design.

All the iVIS units (along with all interconnecting cables of a set are packed in a master box. All the iVIS units first wrapped in a anti-static bag & then will be packed in master box in 2 or 3 layers. Bigger / heavy units shall be packed first (lower layer of box) & than lighter units & all interconnecting cables shall be packed over the lower layer.

4.2 Station List of iVIS

1 set of iVIS shall consist following items;

S/N	Description	Qty
1	Central Switch Unit	01
2	User Unit	As Required
3	QWERTY Keyboard	01 No.
4	Doc Station Unit	As required
5	Handheld Radio	As required
6	Antenna Unit	As required
7	Headsets	As per number of UUs
8	Input Power Cable	01 No.
10	Data Power Cable	As per number of UUs
11	Radio Cable	As required
12	Antenna Cable	As required
13	Doc Station Power Cable	As required
14	Headset Cable	As per number of UUs

Table 8: iVIS – Station List

Note: Refer list of deliverables provided with the iVIS system.

4.3 Unpacking of Material

Unpacking of iVIS also is of very conventional method and there is no special tool required for unpacking. Following are the general guidelines for unpacking of the box;

- a) Place the box on flat surface/ground with its direction upward.
- b) Cut/Tear-off the packing tape paste on the middle of the top side of packing box.
- c) Once the box is opened from the top, remove the packing material from the top. Items packed on the top layers will be visible. Take out all the items & keep them aside.
- d) Once all items are taken out, remove the packing article of down Layer. Take out all the items from this layer also & keep them also aside.
- e) Follow the same procedure, if there is one more layer.
- f) All the units which are taken out from packing box are wrapped in antistatic bags. Unwrap all the items from antistatic bags.
- g) The packing box should be preserved for future use, hence keep all the packing material inside the box again (including antistatic bags) for future re-packing for transportation. Preserve all antistatic bags also.

Now iVIS system is ready for installation and preliminary checks. Refer chapter 5 for installation & preliminary test.

5 Installation

5.1 Important Note for Installation

- a) During factory testing, all the iVIS units had been configured to work as integral Intercom System for one vehicle. Inter-mixing of the units from multiple boxes may hamper the functionality or performance.
- b) Since iVIS is an IP based system so during testing at factory, an unique MAC Id & IP address has been defined to CSU & every User Unit.
- c) User Units are identical hardware & software wise.
- d) The difference between all above mentioned User Units shall be only access control based upon their operational requirement. Access control is configurable which can be configure as per requirement.
- e) Based upon the profile of the User Units, name / short names of User Units also can be configured (Factory Configurable Only). For example, for an armoured vehicle, User Units can be named as follows;
 - Commander Unit - CMDR Unit
 - Section Commander Unit - S-CMDR Unit
 - Section 2IC Unit - SEC-2IC Unit
 - Gunner Unit - GUNR Unit
 - Driver Unit. - DRVR Unit
- f) After unpacking of the packaging box, all the units of iVIS must be installed, fitted in its respective pre-designated locations/slots in the vehicle.
- g) All the components must be connected using respective connecting cables provided and marked.
- h) Start installation, connect various units as shown in chapter 5.2.1 & explained in chapter 5.2.2.

5.2 Installation Steps

5.2.1 Connection Diagram

All the components of iVIS are connected using various connecting cables provided with the system. Figure 22 shows the interconnections between all the units of iVIS.

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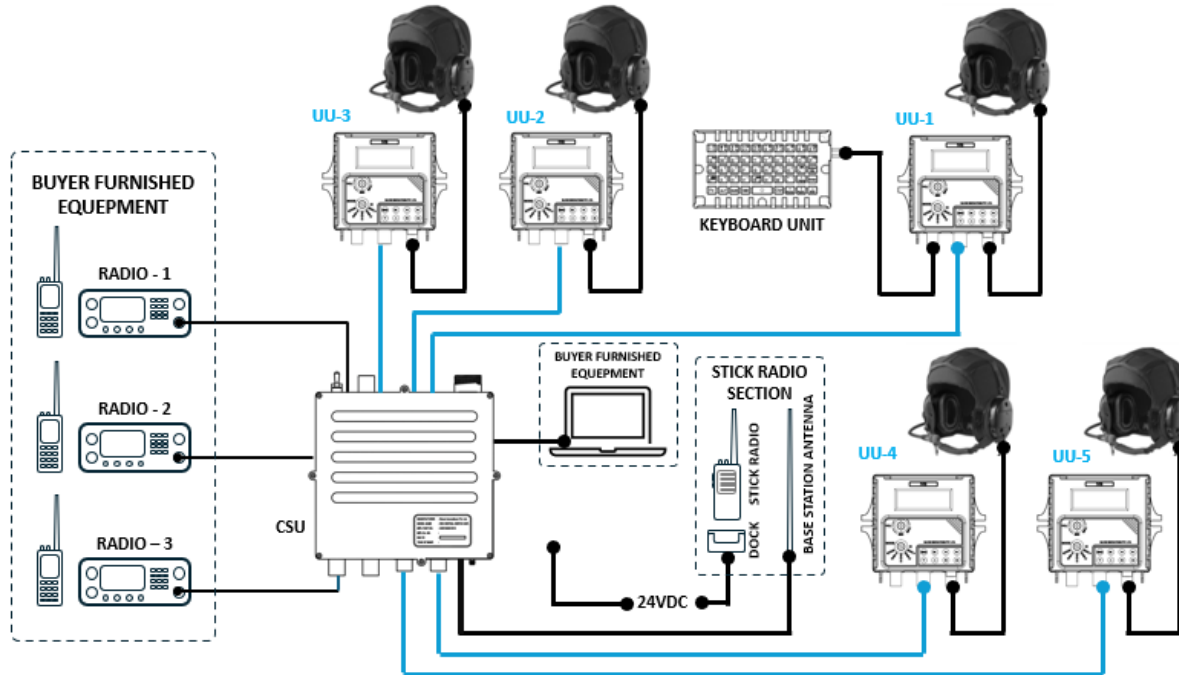


Figure 22: Connection Diagram of iVIS

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5.2.2 Connection Details

To install the iVIS, follow the steps shown in Table 9.

Step	Unit under Installation	Connection Description	Connection Detail	Cable Used
A	CSU – Central Switching Unit			
A1	Input Power 24Vdc to CSU	Keep the ON/OFF switch in OFF position.	Connector 24V-DC of CSU to Vehicle Power. One end of the power cable will be open ended to connect with vehicle power.	CBL11006
		Connect 24V dc (vehicle power) to the connector 24VDC of CSU.		
A2	Connecting Radio Sets to CSU	Place the Radio Sets (provided by customer) at desired location.	Connect audio Connector of 1st Radio Set to R/A-1 of CSU.	CBL21009
		Connect the desired number of Radios (max 04) one by one to the CSU. Desired number of Radio cables are provided with iVIS.	Connect audio Connector of all other Radio Sets to R/A-2, R/A-3 & R/A-4 of CSU respectively.	
		Note: Radio Cables are dependent on the type of Radio sets, provided by customer.		
A3	Connect Base Station Antenna to CSU	Fix the Base Station Antenna at desired location.	One end of the cable to Base Station Antenna & another end to be connected to the connector ANT of CSU.	CBL41001
		1 No. Antenna cable is provided with iVIS.		

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Step	Unit under Installation	Connection Description	Connection Detail	Cable Used
A4	Connect Doc Station to 24V power source	Fix the Doc Station at desired location.	Connect DOC Station power connector to input power source (24Vdc) of Vehicle. One end of the power cable will be open ended to connect with vehicle power.	CBL11006
		Connect the power 24V-DC to DoC Station from vehicle power source.		
B	User Units (UU) – As required			
B1	Install 1 st User Unit UU-1	Identify the Commander Unit either by the marking or just switch on the unit as stand alone. Display will show the type of unit.	Connect the Data Power Cable from ETH of UU to J1 of CSU & observe display.	CBL21051
B2	Input power & data connection to UU-1	Data & power will be given to UU-1 through its Data & power interface (ETH) to the CSU.	Connector ETH of UU1 to be connected to J1 of CSU.	CBL21051
B3	Connect Headset to UU-1	Headset (CPA31004) to be connected to UU-1.	One end of the Headset cable to be connected to the connector HEADSET of UU-1 & another end to be connected to headset.	CBL21052
B4	Identify other User Units	Identify other User Units as explained in Step B1 & fix the unit at desired location accordingly.	-	-
B5	Input power & data connection to rest 4 UUs.	Like Step B2, data & power has to be given to UU-2 to UU-5.	Connector ETH of UU2 to be connected to J2 of CSU. ETH of UU3 to J3 & so on. (Do not use J6 connector of CSU).	CBL21051

Step	Unit under Installation	Connection Description	Connection Detail	Cable Used
B6	Connect Headset to UU-2 to UU-5	Like Step B3, one headset has to be connected to rest of the UUs also.	One end of the Headset cable to be connected to the connector HEADSET of respective UU & another end to be connected to the headset.	CBL21052
B7	Connect QWERTY Keyboard to Commander UU-1	Keyboard Unit can be connected to any of the UU. Connect the Keyboard cable (built-in with Keyboard) to the UU-1.	Connector the keyboard cable to the connector KBRD of UU-1.	-
B8	Set the Mode 'IC' on all UUs	Let all the User Units set to Internal Call.	-	-

Table 9: iVIS Interconnection Details

Once all the interconnections are made as explained in chapter 5.2.2, iVIS is ready to work.

Figure 22 shows generic interconnections between all iVIS units. Figure 23, Figure 24, Figure 25 & Figure 26 shows the interconnections of iVIS for CMTV, AATV, BMP 2K & BMP 2 respectively. Table 10, Table 11, Table 12, Table 13 shows the deliverable list for CMTV, AATV, BMP 2K & BMP 2 respectively.

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Figure 23: Connection Diagram for CMTV

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SN	DESCRIPTION	ITEM ID	QTY
1	iVIS CENTRAL SWITCH UNIT	UCZN10331/01H	1
2	iVIS USER UNIT	UCZN10331/14J	4
3	Headset Cable	CBL21052/0200	4
4	ETH & Power Cable (CMT-CSU to CDR)	CBL21051/0060	1
5	ETH & Power Cable (CMT-CSU to CO-DRV)	CBL21051/0800	1
6	ETH & Power Cable (CMT-CSU to DRV)	CBL21051/1000	1
7	ETH & Power Cable (CMT-CSU to IO)	CBL21051/0550	1
8	ANR Headsets	CPA31004	4
9	Stick Radio with Battery	UCZR10340HH/1	1
10	Stick Radio Dock Station	UCZR10340CS/1	1
11	Vehicle Mount Antenna	UCZR10340VA/1	1
12	Antenna Cable-CMT	CBL41001/0600	1
13	QWERTY Keyboard	CPA41014	1
14	Power Cable (CMT-Veh PWR to CSU)	CBL11006/0750	1
15	Power Cable (CMT-Veh PWR to Docking Station)	CBL11006/0800	1
16	Radio Cable CNR900	CBL21009/0150	2

Table 10: Deliverable List CMTV Vehicle

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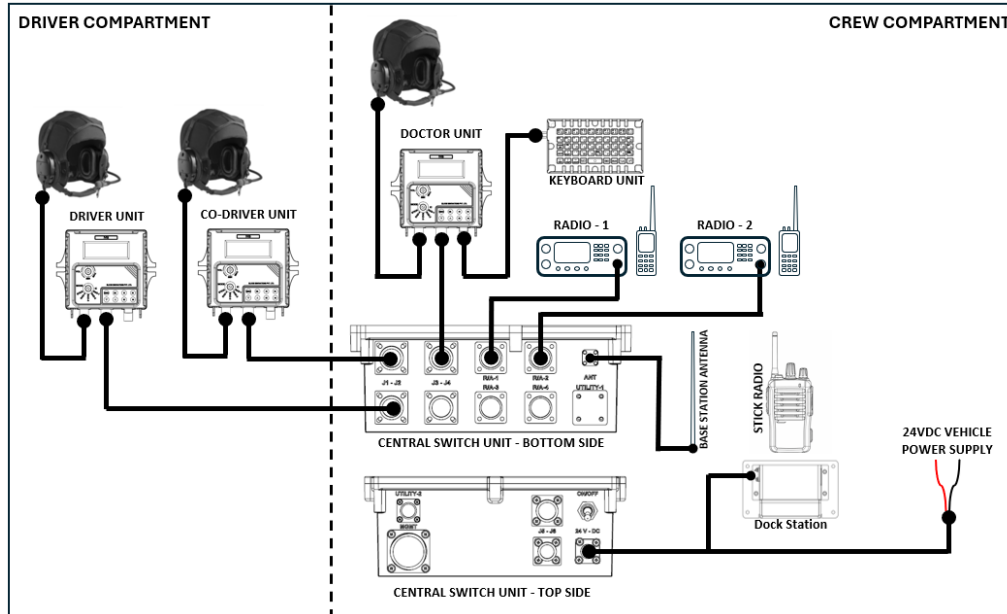


Figure 24: Connection Diagram For AATV

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SN	DESCRIPTION	ITEM ID	QTY
1	CSU	UCZN10331/01H	1
2	UU	UCZN10331/14J	3
3	Headset Cable	CBL21052/0200	3
4	ETH & Power Cable (AAT-CSU to CO-DRV)	CBL21051/0060	1
5	ETH & Power Cable (AAT-CSU to DRV)	CBL21051/0250	1
6	ETH & Power Cable (AAT-CSU to DOC)	CBL21051/0850	1
7	ANR Headsets	CPA31004	3
8	Stick Radio with Battery	UCZR10340HH/1	1
9	Stick Radio Dock Station	UCZR10340CS/1	1
10	Vehicle Mount Antenna	UCZR10340VA/1	1
11	Antenna Cable-AAT	CBL41001/0750	1
12	QWERTY Keyboard	CPA41014	1
13	Power Cable (AAT-Veh PWR to CSU)	CBL11006/0150	1
14	Radio Cable CNR900	CBL21009/0150	2
15	Power Cable (AAT-Veh PWR to Docking Station)	CBL11006/0800	1

Table 11: Deliverable List AATV Vehicle

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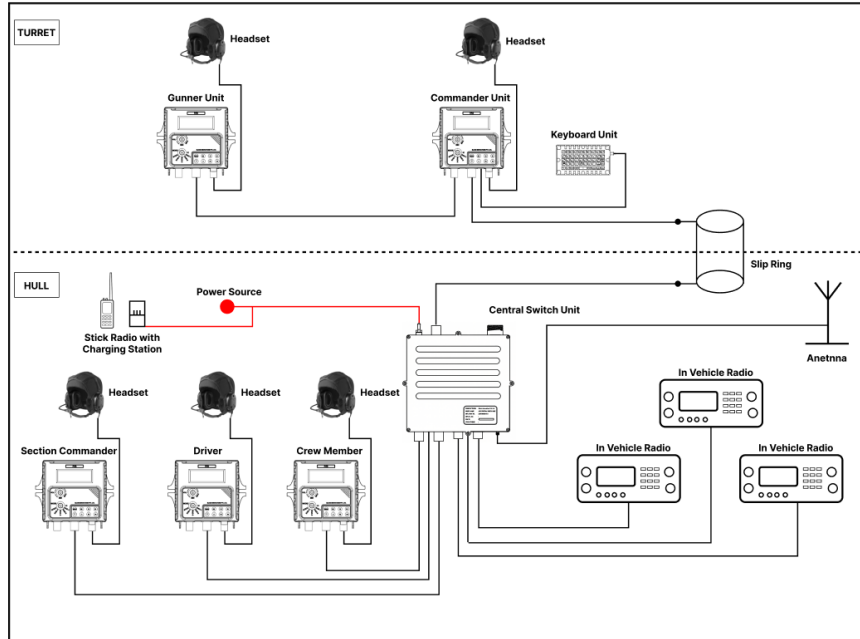


Figure 25: Connection Diagram For BMP 2K

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SN	DESCRIPTION	ITEM ID	QTY
1	iVIS CENTRAL SWITCH UNIT	UCZN10331/01H	1
2	iVIS USER UNIT	UCZN10331/14J	5
3	Headset Cable	CBL21052/0200	5
4	ETH & Power Cable (CSU to CDR)	CBL21051/0060	1
5	ETH & Power Cable (CSU to CO-DRV)	CBL21051/0800	1
6	ETH & Power Cable (CSU to DRV)	CBL21051/1000	1
7	SPE Cable (CSU to SLIP Ring and SLIP Ring to UU)	CBL21058/0650	2
8	ETH & Power Cable (CSU to S2IC)	CBL21051/0550	1
9	ANR Headsets	CPA31004/0200	5
10	Stick Radio with Battery	UCZR10340HH/1	1
11	Stick Radio Dock Station	UCZR10340CS/1	1
12	Vehicle Mount Antenna	UCZR10340VA/1	1
13	Antenna Cable-CMT	CBL41001/0600	1
14	QWERTY Keyboard	CPA41014	1
15	Power Cable (Veh PWR to CSU)	CBL11006/0750	1
16	Power Cable (Veh PWR to Docking Station)	CBL11006/0800	1
17	Radio Cable CNR900	CBL21009/0400	2

Table 12: Deliverable List **BMP 2K** Vehicle

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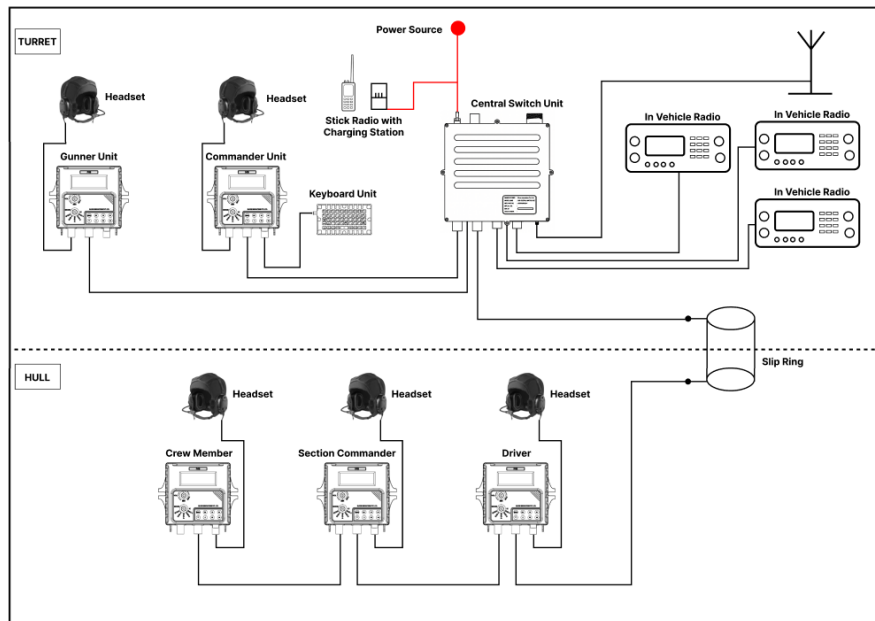


Figure 26: Connection Diagram For BMP 2

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SN	DESCRIPTION	ITEM ID	QTY
1	iVIS CENTRAL SWITCH UNIT	UCZN10331/01H	1
2	iVIS USER UNIT	UCZN10331/14J	5
3	Headset Cable	CBL21052/0200	5
4	ETH & Power Cable (CSU to CDR)	CBL21051/0060	1
5	ETH & Power Cable (CSU to CO-DRV)	CBL21051/0800	1
6	ETH & Power Cable (CSU to DRV)	CBL21051/1000	1
7	SPE Cable (CSU to SLIP Ring and SLIP Ring to UU)	CBL21058/0650	2
8	ETH & Power Cable (CSU to S2IC)	CBL21051/0550	1
9	ANR Headsets	CPA31004/0200	5
10	Stick Radio with Battery	UCZR10340HH/1	1
11	Stick Radio Dock Station	UCZR10340CS/1	1
12	Vehicle Mount Antenna	UCZR10340VA/1	1
13	Antenna Cable-CMT	CBL41001/0600	1
14	QWERTY Keyboard	CPA41014	1
15	Power Cable (Veh PWR to CSU)	CBL11006/0750	1
16	Power Cable (Veh PWR to Docking Station)	CBL11006/0800	1
17	Radio Cable CNR900	CBL21009/0400	2

Table 13: Deliverable List **BMP 2** Vehicle

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5.3 Powering-ON iVIS

Once all the iVIS components are installed and interconnected, the system will be ready for preliminary check. Follow the steps for preliminary check of iVIS;

- Power-on the iVIS system by turning on the vehicle battery power followed by ON/OFF switch of CSU to ON position.
- After few seconds (approx. 20 sec), all the User Units shall come to Idle Mode.
- Observe the display content of the User Units. Figure 27 shows the display content of all the user units in idle mode.

	Unit Type		
Row 1	CMDR	CH:00	Selected Mode
Row 2		VL:00	Volume Level
Row 3		INTCOM	Listen – In Channel
Row 4	<ALARM>		

Figure 27: User Units in Idle State

- From any/all of the headsets, change the Mode knob or Vol knob. You should be able to listen to voice prompt of respective selected channel.

The iVIS system is now ready to use.

6 Operations

6.1 Operation Modes

iVIS system has following 3 modes of operation. The description of all 3 available modes is as follows;

- a) **Intercom Call Mode** – By default all users (User Units) can talk with each other in conference mode. All the interfaced users whose mode is set to **IC** will be able to communicate to each other.
- b) **Radio Mode** - iVIS has a provision to interface with multiple Radio which remain connected to CSU. Any of the User Units can communicate to any desired Radio set in half duplex by selecting the **MODE** knob to desired Radio i.e. RAD-1, RAD-2 & soon.

Multiple Users can select same Radio. Users will be able to communicate with selected Radio in half duplex communication. In this scenario, users will not be able to talk to each other.

- c) **STICK Mode** - iVIS has optional Base Radio Station within CSU to provide the communication with dismounted soldier. If equipped, User Units whose mode is set to **STICK** shall be able to have a half-duplex communication with dismounted soldier.

6.2 Operating Instructions

Commander, and any other User Unit has Mode selection knobs. By changing the knob position, user can select the required mode of operation.

Knob change action is possible only when PTT is in released state. Whenever knob change is initiated, the user will hear a voice prompt to indicate the knob change action has happened. Knob change action takes approx. 1 second to detect the knob change operation and to take desired action.

6.2.1 Call Scenarios on User Unit/s

6.2.1.1 Intercom Communication – *from any of the UU*

Step	Function	Action	Result
1	To communicate with all other internal users	Set MODE knob to 'IC' & Headset PPT is OFF.	User shall be able to listen the voice (in his headset), coming from all other users whose mode is set to 'INC'.
2	To Speak out	Set the Headset PTT Momentary in ON position	User shall be able to transmit the voice.
3	To speak & listen without PTT	Set the Headset PTT is latched in rear ward.	User shall be able to transmit & receive the voice without any further action.
4	To increase the receive volume level	Rotate VOL key in clock wise direction	Headset Volume level (Receive) will be increased.
5	To decrease the receive volume level	Rotate VOL key in anticlock wise direction	Headset Volume level (Receive) will be decreased.

Note – In Intercom mode, if configured User will be able to listen to configured Radio's receive as well as stick radios receive voice

Table 14: Intercom Call

6.2.1.2 Communication with Radio Sets

Step	Function	Action	Result
1	To communicate with desired Radio Set i.g. Radio-1	Set MODE to RAD-1 & Headset PPT is OFF.	User shall be able to listen the voice (in his headset), coming from Radio-1.
2	To Speak out	Set the Headset PTT Momentary in ON position	User shall be able to transmit the voice to RADIO-1.
3	To increase the receive volume level	Rotate VOL key in clock wise direction	Headset Volume level (Receive) will be increased.

Step	Function	Action	Result
4	To decrease the receive volume level	Rotate VOL key in anticlock wise direction	Headset Volume level (Receive) will be decreased.
5	To communicate with any other Radio Set	Set the Mode knob to desired RAD & follow the steps 2 to 4.	Communication will desired radio set will take place.

Table 15: Radio Call

Note: While communicating with any of the Radio Set, Headset PTT shall not be latched to rear ward.

6.2.1.3 Communication with STICK Raio

Step	Function	Action	Result
1	To communicate with STICK Radio Set	Set MODE to STICK & Headset PPT is OFF.	User shall be able to listen the voice (in his headset), coming from STICK Radio.
2	To Speak out	Set the Headset PTT Momentary in ON position	User shall be able to transmit the voice to STICK Radio.
3	To increase the receive volume level	Rotate VOL key in clock wise direction	Headset Volume level (Receive) will be increased.
4	To decrease the receive volume level	Rotate VOL key in anticlock wise direction	Headset Volume level (Receive) will be decreased.

Table 16: Call with STICK Radio

Note: While communicating with STICK Radio, Headset PTT shall not be latched to rear ward.

7 Configuration

Various parameters of iVIS system (including access control) are preconfigured in factory to predefined configuration however these parameters can be reconfigured as per requirement.

All user specific parameters can be set/view by all users however there are certain critical parameters which requires a valid password to set / view. All the configuration parameters are categorized in below mentioned options.

System Info
System Status
Functionality
Settings
Diagnostics
Maintenance

All such critical parameters which require password, are mentioned in Figure 32. Refer chapter 7.1 to get into the configuration mode.

7.1 Get into Configuration Mode

While User Unit (any user) is in idle state, user has to press **MENU** key on built-in keypad. Following options shall be displayed on the display of User Unit.

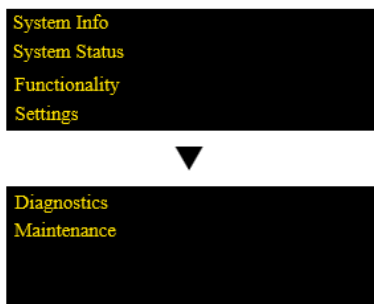


Figure 28: MENU Options

Display will show first 4 options as shown in Figure 28, Press **Down** arrow key to see next 2 options i.e. **Diagnostics & Maintenance**.

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Use Arrow keys to go to desired option. Press **OK** to select the desired option. Now User can select desired option one by one & can view or configure the desired parameters as per requirements.

Figure 29 shows a quick view of all the configuration parameters which can be viewed / configured as per requirements.

Section 7.2 onwards illustrates all the configuration options in detail one by one.

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MENU

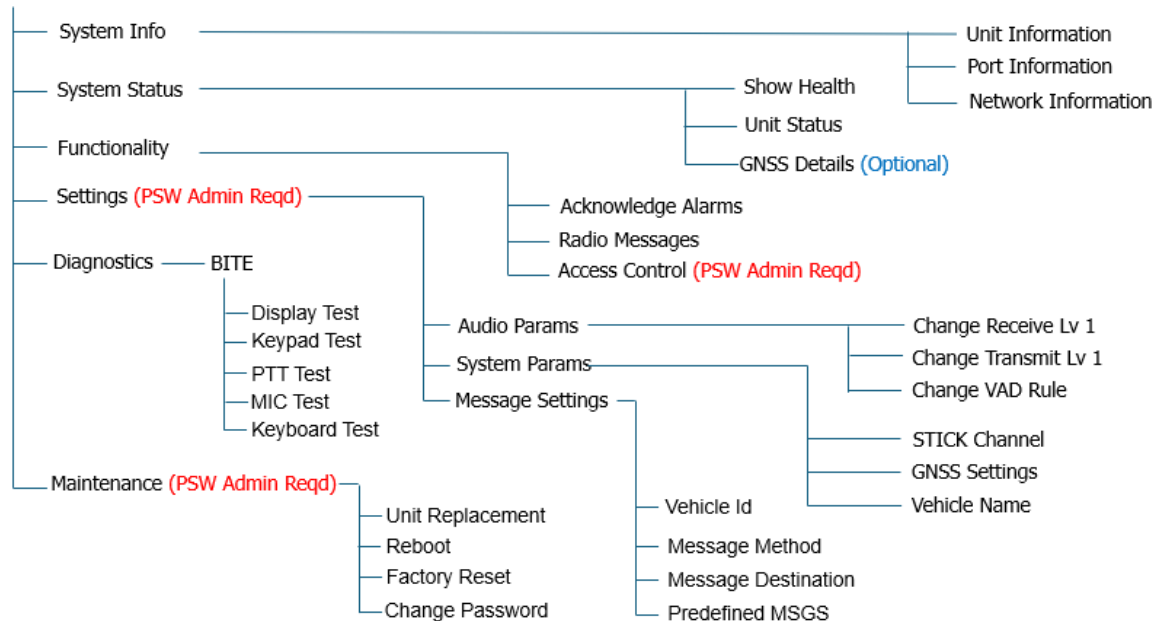


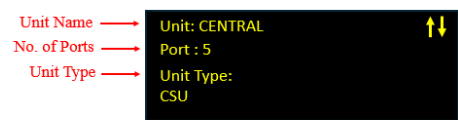
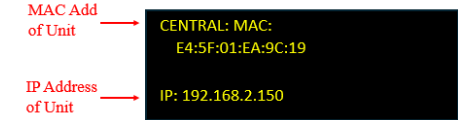
Figure 29: Configuration Tree in a Glance

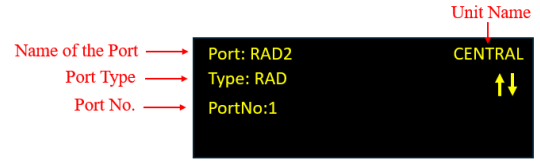
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7.2 Starting Configuration

7.2.1 System Info

System Info can be viewed by any type of User. System Info shall display Unit Information, Port Information & Network Information as shown in Table 17. Follow the steps to view the System Information.

S/N	Action	Action Description	Display Content
A	MENU > System Info > Unit Information	- Display will start displaying the Unit details of every unit (including CSU) one by one; - Unit : Name of the Unit, i.e. CENTRAL or User Unit - Ports : No. of Ports available on this Unit; CSU : No. of Ports as equipped User Units: 01 No. of Port - Unit Type: CSU or CMDR or SCMDR, etc. - Press OK to view the MAC address & IP address of the unit. - Press Down arrow to view the unit info of next unit. - Press MENU key to go back. Note: 'Unit Name' will be displayed as configured.	 Unit Name → Unit: CENTRAL No. of Ports → Port : 5 Unit Type → Unit Type: CSU OK  MAC Add of Unit → CENTRAL: MAC: E4:5F:01:EA:9C:19 IP Address of Unit → IP: 192.168.2.150

S/N	Action	Action Description	Display Content
B	MENU > System Info > Port Information	- Display will start displaying the details of every port, available on various Units. The content details are as follows; Unit Name : Name of the Unit (1st row, right side) Port : Name of the port (1st row, left side) Port Type : RAD, HEADSET, ALRM, FT, etc – (2nd row, left side) Port No. : For CSU, 0 / 1 / 2 / 3 : RAD-1,RAD-2,RAD-3,RAD-4 4 : STICK 5 to 12 : Alarms (if equipped) 13 : For Field Telephone (if equipped) For User Units – Only 1 port i.e. 0 for every type of UU. - Port details will be displayed for every unit one by one. - Press MENU key to go back. Note: ‘Port Name & Port Type’ will be displayed as configured.	 The screenshot shows a black background with yellow text. On the left, three red arrows point to the text: 'Name of the Port' points to 'Port: RAD2', 'Port Type' points to 'Type: RAD', and 'Port No.' points to 'PortNo:1'. On the right, 'Unit Name' points to 'CENTRAL' with a red arrow. Below 'CENTRAL' are two yellow arrows, one pointing up and one pointing down.

S/N	Action	Action Description	Display Content
C	MENU > System Info > Network Information	- Display will start displaying the network parameters of the Unit; Static IP: 192.168.2.102 Sub Netmask: 255.255.255.0 Gateway IP: 192.168.2.254 Mac Id: aa:bb:cc:dd:ee:ff - First Static IP & Subnet Mask will be displayed. - Press Down arrow or OK to view the Gateway IP & MAC Id. - Press MENU key to go back.	IP Address → Static IP: 192.168.2.102 Subnet Mask → Sub Netmask: 255.255.255.0 ▼ / OK Gateway IP → Gateway IP: 192.168.2.254 MAC Add → MAC Addr: aa:bb:cc:dd:ee:ff

Table 17: System Info

7.2.2 System Status

System Status can be viewed by a user of User Units. System Status shall display Health of the Units, Status of various ports & GNSS details as shown in Table 18. Follow the steps to view the System Status.

S/N	Action	Action Description	Display Content
A	MENU > System Status > Show Health	- Display will start displaying the Unit Health one by one; Row 1: Unit Name Row 2: Unit Status, 'UP' or 'DW'. Row 3: Configuration Rev. (required by engineers) Row 4: Connection Info Rev. (required by engineers). - Press Down arrow to view the Health of next unit. - Press MENU key to go back. Note: 'Unit Name' will be displayed as configured.	The diagram illustrates the sequence of unit status displays. It shows three screens, each with a black background and yellow text. The first screen displays 'Unit: CENTRAL', 'Status: DW', 'Config Rev: 0', and 'CM Rev: 0'. The second screen displays 'Unit: CENTRAL', 'Status: DW', 'Config Rev: 0', and 'CM Rev: 0'. The third screen displays 'Unit: CMDR', 'Status: UP', 'Config Rev: 0', and 'CM Rev: 1'. Arrows indicate the flow: 'Status of Unit' points to the first screen, 'SW Revisions' points to the second screen, and a downward arrow points from the second screen to the third screen.

S/N	Action	Action Description	Display Content
B	MENU > System Status > Unit Status	- This option is applicable only on User Units to view the listen-in rights of same unit only. - It also shows the outgoing voice settings when PTT1 / PTT2 is pressed. - Display will start showing the details as follows; Row 1 : Name of the port as configured. Row 2 : Left Side – Selected Channel; 00,01, & so on for IC, RAD1, RAD2, etc. Row 2 : Right Side – Position of Vol knob Row 3 : NRML means Tx & Rx both Row 4 : Not Used. - Press OK to view next screen which will show listen-in rights of IC, RAD1, RAD2 for this unit. 'X' will be marked for all such source whose voice can be listen-in by this unit. - Press OK for next screen which will show the listen-in rights of FT (Field Telephone) for this unit. - Press OK for next screen which will show the configured settings of PTT1 & PTT2. For example, 'IC' for PTT1 & 'NC' for PTT2. It means, when the user of this unit, keeps the PTT1 in ON position, his own voice will be transmitted to IC (provided defined access control) & PTT2 is not configured.	Port Name → Selected Mode → Port:CMD CH:05 NRML VL:05 ↑↓ ← Selected Vol Level OK IC ---X--- RAD-1 ---X--- RAD-2 ----- OK RAD-3 ----- RAD-4 ----- RAD-5 ---X--- OK FT ---X--- OK PTT1 IC PTT2 NC OK

S/N	Action	Action Description	Display Content
C	MENU > System Status > GNSS Details*	This is to view the Global position of the User Unit. ▪ GNSS is the optional feature so global positioning coordinates will be displayed only if the unit is equipped for this feature. ▪ If unit is not configured for GNSS, display will be show as per Fig 1. ▪ If GNSS is configured but not connected display will be show as per Fig 2. ▪ If GNSS is configured and connected. display will show Lat/Long/Elevation and current time sync with the GNSS/ALNS as shown in figure 3 & 4.	① GSS not Enabled Press MENU to Exit ② GSS not Connected Press MENU to Exit ③ Lat: +0.1011000 Lon: +0.0024167 14:32:05 15-Jun-2026 ④ Eltvn: 0.00 m Speed: 10.00 km/hr 14:32:05 15-Jun-2026

Table 18: System Status

7.2.3 Functionality

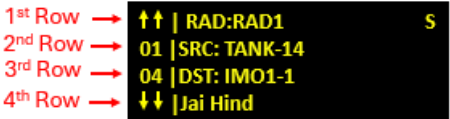
Through 'Functionality', user can perform following actions;

- Acknowledge alarms - To acknowledge the alarms (if equipped and enabled)
- Radio Messages - To send or receive messages through Radio
- Access Control - To set the Radio Parameters & Access Control of Radios

S/N	Action	Action Description	Display Content
A	MENU > Functionality > Acknowledge Alarms	iVIS has multiple alarm input ports as optional. If equipped and enabled and if any of the alarm port has detected an input, iVIS shall generate a voice prompt/Beep to desired users in their headsets. Voice Prompt/Beep will be switched off when alarm is acknowledged. In case an alarm is displayed, press OK to acknowledge followed by MENU to go back. If no alarms configured screen will show Fig 1 content.	① No Alarms To Display ② Port: AL1 NRML Ack: NRML MENU OK ↑↓ ③ Port: AL1 NRML Ack: NRML MENU OK ↑↓

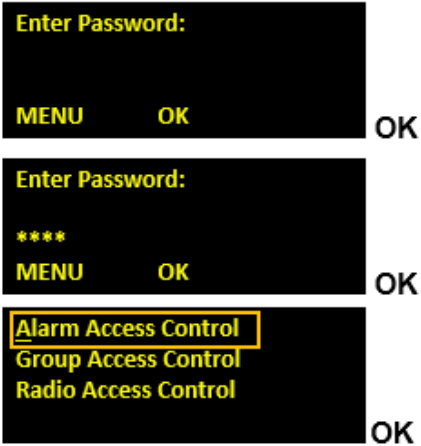
S/N	Action	Action Description	Display Content
		If alarms are configured, to acknowledge alarm follow the steps as per Fig 2, Fig 3, Fig 4, Fig 5, Fig 6. Up/Down arrow navigate to respective alarm, press OK to select the alarm. Press Up/Down arrow to navigate to Acknowledgement method, press OK to Acknowledge the alarm. ACK action will acknowledge the selected alarm and no prompts will be played in the headgear, till next occurrence of the same alarm PRMNT ACK will permanently acknowledge the selected alarm for the current user. No prompt will be played in the headgear of current user even after reactivation of alarm.	4 Port: AL1 NRML Ack: NRML MENU OK ↑↓ 5 Port: AL1 NRML Ack: NRML MENU OK ↑↓ 6 Confirm Ack Change ? MENU OK

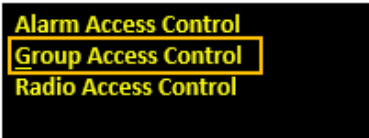
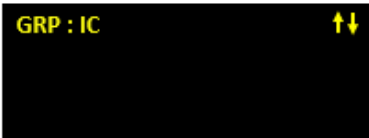
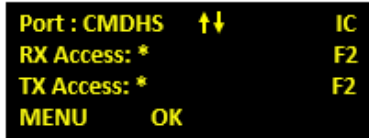
S/N	Action	Action Description	Display Content
B1	MENU > Functionality > Radio Messages > OutBox	- Display will show 'OutBox', 'Inbox', 'Compose' & 'Pre Defined'. - Select 'OutBox' to view the SENT msgs. 1st Row - shows UP arrow & radio through which is used to send the message. Press UP arrow key to view previous msg. 'S' will be shown on the right corner of the 1st row, if msg has been sent to desired destination. 2nd Row – shows number of msg & source of the msg. 3rd Row – shows total number msgs in OutBox & destination of the msg. 4th Row – shows Down arrow & the msg content. To view next sent msg, press Down arrow key. A total of 99 msgs can be saved in OutBox. While viewing a msg, press F2 to delete the msg.	 OK  1st Row → ↑↑ RAD:RAD1 S 2nd Row → 01 SRC: TANK-14 3rd Row → 04 DST: IMO1-1 4th Row → ↓↓ Jai Hind

S/N	Action	Action Description	Display Content
B2	MENU > Functionality > Radio Messages > Inbox	- Select Inbox to view the incoming msgs. - Msgs from Inbox also can be viewed as explained for OutBox. - While viewing a msg, press F2 to delete the msg.	 
B3	MENU > Functionality > Radio Messages > Compose	- Select 'Compose' to compose a msg to send to someone. - Display content will be as shown in adjacent picture. User can compose desired msg as explained in the picture. Msg content can have max 38 characters. - Qwerty Keyboard, connected to the User Unit also can be used to compose msg. - Msg can be composed through functional keypad also which is available on the top of the user unit.	

S/N	Action	Action Description	Display Content
		- From built-in functional keypad, following 4-character sets are supported; - CAPS English characters - Small English characters - Numbers - Special characters - User can select the desired character set by pressing F4 key. - After selecting the desired character set, desired character can be entered by pressing UP/Down arrow keys and use function keys to manage the text). - After entering the desired text message, press OK. - User will be prompted to select the desired Radio & Destination as shown in adjacent picture.. - Press OK & msg will be saved to send it to desired destination. - iVIS shall transmit the saved msg to desired destination in the background & sent msg will be saved in OutBox with a suffix 'S' as explained earlier. - Press MENU key to go back.	The diagram illustrates the user interface for character selection and message sending. The top screen shows a character set selection menu with options: ABC, 01/38, F1 <, and ->F3. The current position of the cursor is indicated by a red arrow pointing to the 'ABC' option. The selected character set is 'ABC'. The screen also displays 'Min 3' and 'F2 Del'. The bottom screen shows the 'Radio Port' selection screen with 'Port: RAD2' and 'Destination' selection screen with 'Dest: IMOD-1'. The status bar at the bottom indicates 'Status: Msg Queued'.

S/N	Action	Action Description	Display Content
B4	MENU > Functionality > Radio Messages > Pre Defined	- iVIS system has a bank of 10 No. predefined msgs in which 10 msgs can be stored in advance which will be used frequently. - Select Pre Defined to view the pre defined msgs. 1st Row - shows UP arrow along with 1st pre-defined msgs. 2nd Row – shows msg number. 3rd Row – shows total number of pre-defined msgs. 4th Row – shows Down arrow key. - To view next pre-defined msg, press Down arrow key. - While viewing a pre-defined msg, press OK to send the msg. - User will be prompted to select the desired Radio & Destination as shown in adjacent picture.. - Press OK & msg will be saved to send it to desired destination. - Press MENU key to go back. - Pre defined msgs can be edited also, refer chapter 7.2.4.	OK 1st Row → ↑↑ Jai Hind 2nd Row → 01 3rd Row → 10 4th Row → ↓↓ OK Radio Port → Port: RAD2 ↑↓ Destination → Dest: IMOD-1 OK Status: Msg Queued

S/N	Action	Action Description	Display Content
C1	MENU > Functionality > Access Control > Alarm Access Control	▪ After selecting Access Control, user shall be prompted to enter the password. ▪ On entering the correct password, 3 options shall be displayed. ▪ 1st option is Alarm Access Control. ▪ This option is to configure the access control of alarms. ▪ Press MENU to go back.	 The display content consists of three sequential screenshots of the device's interface: First Screenshot: A black screen with yellow text. At the top, it says "Enter Password:". Below this, there are two options: "MENU" and "OK". To the right of the "OK" option, the text "OK" is displayed. Second Screenshot: A black screen with yellow text. At the top, it says "Enter Password:". Below this, there are four asterisks "****" indicating the password entry field. Below the asterisks, there are two options: "MENU" and "OK". To the right of the "OK" option, the text "OK" is displayed. Third Screenshot: A black screen with yellow text. It shows a menu with three options: "Alarm Access Control", "Group Access Control", and "Radio Access Control". The "Alarm Access Control" option is highlighted with a yellow rectangular border. To the right of the menu, the text "OK" is displayed.

S/N	Action	Action Description	Display Content
C2	MENU > Functionality > Access Control > Group Access Control	2nd option is Group Access Control. This option is to configure the Intercom Group for the receive (Rx) & transmit (Tx) voice of various ports i.e. all user units. - Select the Group Access Control by pressing OK. - All the created groups shall be displayed. For example; adjacent picture is showing group IC. In case of multiple groups, use arrow key to select the desired group & press OK. - Voice port of one of the User Unit will be displayed. By default Rx & Tx both are configured for shown port in the group IC. '**' means enabled. - Key F2 to be pressed multiple times to set / reset the Rx & Tx. - Once Rx/Tx are set/reset as required, press OK to save else press MENU to exit.	 Alarm Access Control <u>Group Access Control</u> Radio Access Control OK  GRP : IC ↑↓ OK  Port : CMDHS ↑↓ IC RX Access: * F2 TX Access: * F2 MENU OK OK

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S/N	Action	Action Description	Display Content
		- Once the Rx/Tx of first user unit are configured, use UP/Down key to go to the next user unit to configure the Rx/Tx of rest of the user units. - All the user units shall be displayed one by one as shown in adjacent picture. - While viewing a user unit, use F2 to set/reset the Rx / Tx of that unit. - Press OK to save. - Press MENU to exit.	Port : SCMDHS ↑↓ IC RX Access: * F2 TX Access: * F2 MENU OK OK Port : S2ICHs ↑↓ IC RX Access: * F2 TX Access: * F2 MENU OK OK Port : DRVHS ↑↓ IC RX Access: * F2 TX Access: * F2 MENU OK OK Port : GNRHS ↑↓ IC RX Access: * F2 TX Access: * F2 MENU OK OK

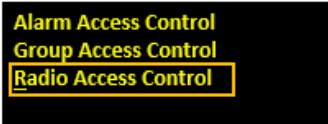
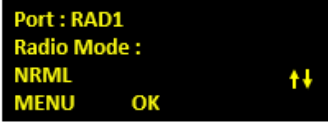
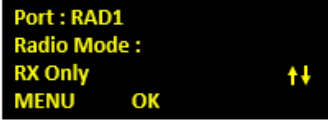
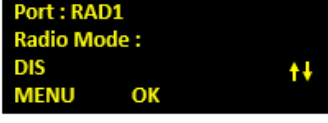
S/N	Action	Action Description	Display Content
C3	MENU > Functionality > Access Control > Radio Access Control	3rd option is Radio Access Control. This option is to configure the receive (Rx) & transmit (Tx) voice of various Radio ports. - Select the Radio Access Control by pressing OK. - Rx & Tx settings of first Radio port will be displayed. By default Rx & Tx both are configured for all the radio ports. - Use Up/Down key to set / reset the Rx & Tx. When Up/Down key is pressed, it will show 3 options i.e.; - RX Only - Only Rx receive voice - DIS - No receive (Rx) & no transmit (Tx) - NRML - Normal means Rx & Tx both - Once Rx/Tx are set/reset as required, press OK to save & press MENU to exit. - Similarly, select next radio port for setting Rx/Tx.	 OK  OK  OK  OK

Table 19: Functionality

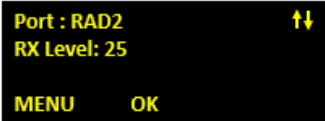
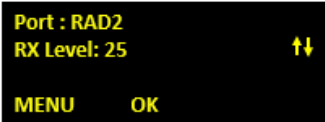
7.2.4 Settings (Require Password)

Through this option, Audio Parameters (Receive / transmit voice level of various units, PTT activation settings), System parameters & message settings can be set / view.

Note : Receive Level is the sense level of Radio ports input to intercom system OR Headset MIC input sense level

Transmit level is Out put level at the radio port of intercom system

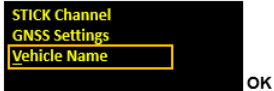
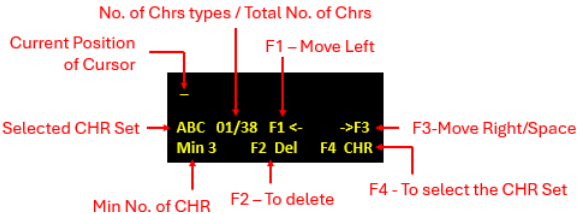
S/N	Action	Action Description	Display Content
A1	MENU>Settings> Audio Params > Change Receive Lv1	- After selecting Settings, user shall be prompted to enter the password. - On entering the correct password, 3 options shall be displayed. 1st option is Audio Params (refer adjacent figure 1). Which is to configure the transmit level, receive level & VAD Rule of Radio ports & STICK radios. - Select Audio Params, figure 2 will be displayed. Use arrow keys to select the desired option & press OK. - On selecting Change Receive Lv1, first radio port will be displayed, refer figure 3.	1 Audio Params System Params Message Settings OK 2 Change Receive Lv 1 Change Transmit Lv 1 Change VAD Rule OK 3 Port : RAD1 RX Level: 19 MENU OK ↑↓ ▼

S/N	Action	Action Description	Display Content
		- Use arrow keys to select the desired radio port & press OK. - Figure 4 shows when Radio 2 is selected. - Use UP arrow key to increase the receive level of Radio 2 & Down arrow key to decrease the receive. Max receive level can be 30 & min 0. - Press OK to save. - Similarly, configure the receive level of any other desired radio. - Press OK to save & press MENU to exit.	4  OK 5  OK

S/N	Action	Action Description	Display Content
A2	MENU>Settings> Audio Params > Change Transmit Lv1	- On selecting Audio Params > Change Transmit Lv1, first radio port will be displayed to configure its transmit level, refer adjacent figure 1 & 2. - Use arrow keys to select the desired radio port & press OK. Figure 3 shows when Radio 2 is selected. - Use UP arrow key to increase the transmit level of Radio 2 & Down arrow key to decrease the transmit level. Max receive level can be 40 & min 0. - Press OK to save. - Similarly, configure the receive level of any other desired radio. - Press OK to save. Press MENU to exit.	① Change Receive Lv 1 Change Transmit Lv 1 Change VAD Rule OK ② Port : RAD1 TX Level: 19 MENU OK ↑↓ ③ Port : RAD2 TX Level: 25 MENU OK ↑↓ ④ Port : RAD2 TX Level: 25 MENU OK ↑↓

S/N	Action	Action Description	Display Content
A3	MENU>Settings> Audio Params > Change VAD Rule	- This is to configure the trigger point of transmit voice of radio ports & user headsets. - On selecting Audio Params > Change VAD Rule, first radio port will be displayed to configure its VAD Rule, refer adjacent figure 1 & 2. - Use arrow keys to select the desired radio port or user & press OK. Desired Radio/User Headset to be selected using arrow keys. Figure 3 shows when Radio2 is selected. - Use arrow key to view all the following options to configure the VAD for selected Radio / User Headset; VAD and PTT Based – VOX + PTT, applicable only on User Headsets. COR Based – Applicable only for Radios. If Radio set is supporting COR signal, that will be used to open the voice path. PTT Based – Voice part will be opened after detecting PTT only. Applicable only for User headsets. VOX Based – Voice path will be opened once voice is detected. Applicable for Radio & User headsets. OUT Disabled – No transmit voice. - While an option is displayed, press OK to save. - Press MENU to exit.	① Change Receive Lv I Change Transmit Lv 1 Change VAD Rule OK ② Port : RAD1 VAD Rule: VAD and PTT Based MENU OK ▼ ③ Port : RAD2 VAD Rule: VAD and PTT Based MENU OK OK ④ Port : RAD2 VAD Rule: VAD and PTT Based MENU OK OK

S/N	Action	Action Description	Display Content
B1	MENU>Settings> System Params > STICK Channel	- After selecting Settings, user shall be prompted to enter the password. - On entering the correct password, 3 options shall be displayed as shown in figure 2.. - Select 2nd option System Params (refer adjacent figure 2). Which is to configure the STICK Channel, GNSS Settings & Vehicle Name as shown in Figure 3. - Select STICK Channel and press OK. - Figure 4 will be displayed wherein channel of STICK Radio need to be selected (1 to 16). - Use Arrow key to select the desired channel & press OK to save. - Press MENU to exit.	① Enter Password: **** MENU OK OK ② Audio Params System Params Message Settings OK ③ STICK Channel GNSS Settings Vehicle Name OK ④ Channel : 1 ↑↓ OK
B2	MENU > Settings > System Params > GNSS Settings	‘GNSS’ is the optional feature so global positioning coordinates will be displayed only if the unit is equipped for this feature.	-

S/N	Action	Action Description	Display Content
B3	MENU > Settings > System Params > Vehicle Name	- This is to define the name to the vehicle. - On selecting Settings > Vehicle Name, display content will be as shown in adjacent picture. - Here vehicle name can be defined. It should be as short as possible. - Qwerty Keyboard (external unit) or functional keypad which is available on the top of the user unit, can be used to define vehicle name. - From built-in functional keypad, following 4-character sets are supported; - CAPS English characters - Small English characters - Numbers - Special characters User can select the desired character set by pressing F4. - After selecting the desired character set, desired character can be entered by pressing UP/Down arrow keys and use function keys to manage the text). - After entering the desired vehicle name, press OK to save & press MENU key to go back.	 The screenshot shows a black screen with yellow text. At the top, it says 'STICK Channel', 'GNSS Settings', and 'Vehicle Name'. A yellow rectangular box highlights the 'Vehicle Name' field. To the right of the screen is an 'OK' button.  The diagram shows a keypad layout with various function keys and their actions: No. of Chrs types / Total No. of Chrs: Points to the '01/38' display. Current Position of Cursor: Points to the cursor position on the 'ABC' character set. F1 - Move Left: Points to the 'F1 <' key. Selected CHR Set: Points to the 'ABC' character set. Min No. of CHR: Points to the 'Min 3' display. F2 - To delete: Points to the 'F2 Del' key. F3 - Move Right/Space: Points to the 'F3 >' key. F4 - To select the CHR Set: Points to the 'F4 CHR' key.

S/N	Action	Action Description	Display Content
C1	MENU > Settings > Message Settings > Vehicle Id	- After selecting Settings, user shall be prompted to enter the password. - On entering the correct password, 3 options shall be displayed as shown in figure 2.. - Select 3rd option Message Settings (refer adjacent figure 2). Which is to configure various parameters shown in figure 3. - Select Vehicle Id and press OK. - Figure 4 will be displayed which shows the Help for setting Vehicle Id. Press OK to set the Vehicle Id. - Enter the Vehicle Id, refer figure 5. - Format of the Vehicle Id is 'Name-SSID' where name can have 3 to 4 characters & SSID can have 0 to 15 characters (alphanumeric). - Though the total length of Vehicle Id can be 21 characters (alpha numeric) however it is recommended to keep the Vehicle Id as short as possible. - After entering the desired Vehicle Id, press OK to save. - Press MENU key to exit.	① Enter Password: **** MENU OK OK ② Audio Params System Params Message Settings OK ③ Vehicle Id Message Method Message Destination Predefined MSGS OK ④ FORMAT: NAME-SSID Name: 3-4 ALPHANUM SSID: 0-15 OK TO CONTINUE OK ⑤ Current Position of Cursor → No. of Chrs types / Total No. of Chrs → F1-Move Left Selected CHR Set → ABC 01/07 F1 <- ->F3 F3-Move Right/Space Min 3 F2 Del F4 CHR Min No. of CHR → F2 - To delete → F4-To select the CHR Set

S/N	Action	Action Description	Display Content
C2	MENU > Settings > Message Settings > Message Method	iVIS is designed to send the text message from the User unit or from external application running on Laptop/PC. The message method field defines who is the source of the message i.e. Intercom User unit or external laptop or PC application. After selecting Settings, user shall be prompted to enter the password. - On entering the correct password, 3 options shall be displayed as shown in figure 2. Select 3rd option Message Settings (refer adjacent figure 2). Which is to configure various parameters shown in figure 3. Select Message Method. - Use arrow keys to select desired port. - While desired port is displayed, press OK, arrow keys will be shifted to 2nd row to configure MSG SRC as shown in figure 5. - Select desired MSG SRC by using arrow key. MSG SRC can be either PC/LAPTOP or INTERCOM. Press OK to save.	① Enter Password: **** MENU OK OK ② Audio Params System Params Message Settings OK ③ Vehicle Id Message Method Message Destination Predefined MSGS OK ④ Port : RAD1 MSG SRC : PC/LAPTOP MSG MODE: SERIAL OK ⑤ Port : RAD1 MSG SRC : PC/LAPTOP MSG MODE: SERIAL OK PC/LAPTOP INTERCOM

S/N	Action	Action Description	Display Content
		- Now arrow keys will be moved to 3rd row to configure MSG MODE as shown in figure 6. Select desired MSG MODE by using arrow key. It can be either SERIAL or VOCIE. Press OK to save. - In case 'SERIAL' is selected as MSG MODE, figure 7 will be displayed to set serial port communication parameters. Communication parameters have to be configured as per the interfaced device. - Select the desired parameter by pressing OK & use arrow keys to set the value as shown in adjacent figures. - Press OK to save & MENU to exit.	⑥ Port : RAD1 MSG SRC : PC/LAPTOP MSG MODE: SERIAL ⑦ BAUD : 9600 DataBit : 8 Parity : None StopBit : 2 ⑧ BAUD : 9600 DataBit : 8 Parity : None StopBit : 2 ⑨ BAUD : 9600 DataBit : 8 Parity : None StopBit : 2 ⑩ BAUD : 9600 DataBit : 8 Parity : None StopBit : 2

S/N	Action	Action Description	Display Content
C31	MENU > Settings > Message Settings > Message Destination > Add Dest	This is to configure the message destination which will be sent through desired radio port. - While getting into MENU > Settings > Message Settings > Message Destination (after entering correct password), figure 2 will be displayed to configure the Message Destination. - First destination to be configured using 'Add Dest' & then destination has to be linked with desired Radio Port through 3rd option 'Select Dest'. - Select 'Add Dest' & press OK, content shown in figure 3 will be shown. This is the help page to configure the destination which shows the format & the length of the message destination. A max of 16 message destination can be configured. - Press OK to configure the message destination, content shown in figure 4 will be displayed. Enter the destination name. - After entering the message destination in correct format, press OK to save. - Press MENU to exit.	① Vehicle Id Message Method Message Destination Predefined MSGS OK ② Add Dest Delete Dest Select Dest Edit Dest OK ③ FORMAT: NAME-SSID Name: 3-6 ALPHANUM SSID: 0-15 OK TO CONTINUE OK ④ A Selected CHR Set → ABC 01/09 F1 <- →F3 F3 - Move Right/Space Min 5 F2 Del F4 CHR F4 - To select the CHR Set Min No. of CHR F2 - To delete

S/N	Action	Action Description	Display Content
C32	MENU > Settings > Message Settings > Message Destination > Delete Dest	- To delete a message destination, select 'Delete Dest' & press OK. - Content shown in figure 2 will be shown. It will show all the configured message destination. - Use arrow keys to select the desired destination. - While displaying a destination, press OK to delete that destination. It will prompt to confirm. Press OK to confirm else press MENU to exit.	① Add Dest Delete Dest Select Dest Edit Dest OK ② ↑↑ PANZER-1 01 09 ↓↓ OK ③ Confirm to delete? MENU OK OK

S/N	Action	Action Description	Display Content
C33	MENU > Settings > Message Settings > Message Destination > Select Dest	This settings is to link a configured message destination with a radio port. - Go to Select Dest & press OK. - Content shown in figure 2 will be shown. It will show the radio port & the destination which is linked with shown radio port. - Use arrow keys to select the desired radio port. While displaying various radio port, linked destination also will be displayed. - To change or select the message destination for a radio port, select desired radio port by pressing OK, content shown in figure 3, will be displayed. - Use arrow key to select the desired message destination. All the configured message destination will be displayed one by one. - While showing desired message destination, press OK. - Selected message destination will get linked with selected radio port. - Press MENU to exit.	① Add Dest Delete Dest Select Dest Edit Dest OK ② Port : RAD1 ↑↓ dest : PANZER-1 OK ③ ↑↑ VEER-15 01 09 ↓↓ OK ④ Status : success OK

S/N	Action	Action Description	Display Content
C34	MENU > Settings > Message Settings > Message Destination > Edit Dest	This setting is to edit a configured message destination. - Go to Edit Dest & press OK. - Content shown in figure 2 will be shown. It will show configured message destination. Use arrow key to view next destination. - While displaying a destination, press OK to edit that destination. Content shown in figure 3 will be displayed. - Edit the message destination as desired & press OK to save. - Press MENU to exit.	① Add Dest Delete Dest Select Dest Edit Dest OK ② ↑↑ PANZER-5 01 09 ↓↓ OK ③ PANZER-5 Selected CHR Set → ABC 01/09 F1 <- ->F3 F3 - Move Right/Space Min 5 F2 Del F4 CHR F4 - To select the CHR Set Current Position of Cursor → No. of Chrs types / Total No. of Chrs → F1 - Move Left Min No. of CHR F2 - To delete

S/N	Action	Action Description	Display Content
C41	MENU > Settings > Message Settings > Predefined Msgs > Add Msg	To eliminate the time consuming in message composition, few predefined messages can be configured which might be used frequently by users. - While getting into MENU > Settings > Message Settings > Predefined MSGS (after entering correct password), figure 2 will be displayed to set the predefined messages. - Select 'Add Msg' & press OK, content shown in figure 3 will be shown. This is to add a predefined msg. - A total of 10 msgs can be defined. - Message length can be 03 to 32 characters. - Figure 3 shows the help to enter the desired message. - Press OK to save the message & press MENU to exit.	① Vehicle Id Message Method Message Destination Predefined MSGS OK ② Add Msg Delete Msg Edit Msg OK ③ A Current Position of Cursor No. of Chrs types / Total No. of Chrs F1 - Move Left Selected CHR Set ABC 00/32 F1 <- -> F3 F3 - Move Right/Space Min 3 Min No. of CHR F2 Del F4 CHR F4 - To select the CHR Set F2 - To delete

S/N	Action	Action Description	Display Content
C42	MENU > Settings > Message Settings > Predefined Msgs > Delete Msg	- While getting into MENU > Settings > Message Settings > Predefined MSGS (after entering correct password), figure 2 will be displayed to set the predefined messages. - Select 'Delete Msg' & press OK, content shown in figure 3 will be shown. This is the list of the predefined messages. - Row 1 will show the message content, - Row 2 will show the message number. - Row 3 will show total number of predefined msgs. - Figure 3 is showing 1st predefined msg out of 9. - Use arrow keys to select desired predefined msg. - While viewing a predefined msg, press OK to delete that predefined msg. - Message will be deleted & success msg will be shown as shown in figure 4. - After few seconds, next predefined message will be shown. User can continue to delete any other predefined msg as per requirement. - Press MENU to exit.	① Vehicle Id Message Method Message Destination Predefined MSGS OK ② Add Msg Delete Msg Edit Msg OK ③ ↑↑ Jai Hind 01 09 ↓↓ OK ④ Status : success OK

S/N	Action	Action Description	Display Content
C43	MENU > Settings > Message Settings > Predefined Msgs > Edit Msg	This setting is to edit a predefine message. - While getting into MENU > Settings > Message Settings > Predefined MSGS (after entering correct password), figure 2 will be displayed to edit the predefined msgs. - Content shown in figure 2 will be shown. It will show all the predefined msg one by one. Use arrow key to view next predefine msg. - While displaying a predefine msg, press OK to edit that msg. Content shown in figure 3 will be displayed. - Edit the message as desired & press OK to save. - User can continue to edit any other predefined msg as per requirement. - Press MENU to exit.	① Vehicle Id Message Method Message Destination Predefined MSGS OK ② Add Msg Delete Msg Edit Msg OK ③ ↑↑ Jai Hind 01 09 ↓↓ OK Current Position of Cursor No. of Chrs types / Total No. of Chrs F1 - Move Left Selected CHR Set → ABC 08/32 F1 ← → F3 ← F3 - Move Right/Space Min 3 F2 Del F4 CHR Min No. of CHR F2 - To delete F4 - To select the CHR Set

Table 20: Settings

7.2.5 Diagnostics (BITE)

Built-in Test Equipment (BITE) refers to integrated hardware and software in systems that perform self-diagnostic tests to detect, locate, and report faults to support maintenance. It enhances safety and reduces downtime. Every type of user shall be able to perform the diagnostics. It must be noted that BITE actions/commands are only to test the hardware which is associated with the unit on which BITE test is being conducted. Follow the steps shown in Table 21.

S/N	Action	Action Description	Display Content
A1	MENU > Diagnostics> Display Test	- When cursor is on Display Test, press OK to go further or press MENU to go back. - When press OK, there will be 3 tests of display test as shown in picture. Press OK to start the testing. Test 1 - will print 'P' in every character. If 'P' is displayed in every character, press OK for Test 2. Test 2 - All characters shall be switched off. If yes, press OK for Test 3. Test 3 - will print '8' in every character. If '8' is displayed in every character, press OK. - If Display test is completed successfully. Press OK to come out from display test. - If any error is observed, report to technician for issue resolution.	① Display Test Keypad Test PTT Test Mic Test ② Disptest in 3 Stage ALL-ON, ALL-OFF, ALL-8 OK to continue MENU to cancel ③ Test 1 ④ Test 2 ⑤ Test 3

S/N	Action	Action Description	Display Content
B	MENU > Diagnostics> Keypad Test	- When cursor is on Keypad Test, press OK to go further or press MENU to go back. Refer picture 1. - When press OK, display will show picture 2. Press OK to start the keypad test else press MENU to go back. - If OK is pressed to start the test, iVIS will ask to press any of the key, available on keypad. Refer picture 3. - If user press F4, iVIS will identify the pressed key & it will be displayed on display. Refer picture 4. - Press next key, iVIS will identify & pressed key name also will be displayed in content. User can test every key except MENU key. - If iVIS identifies all the keys successfully, means Keypad test is passed successfully. - If any error is observed, report to technician for error fixing. - Press MENU key, iVIS will prompt to exit or resume testing. Refer picture 5. - Press OK to stop the testing & MENU to resume the testing.	① Display Test Keypad Test PTT Test Mic Test ② Key Test function OK to continue MENU Key to cancel ③ Press desired key ④ Press desired key Key is F4 ⑤ Halt key test? OK to halt MENU key to resume

S/N	Action	Action Description	Display Content
C	MENU > Diagnostics> PTT Test	- When cursor is on PTT Test, press OK to go further or press MENU to go back. Refer picture 1. When press OK, display will show picture 2.  - Headset has a PTT lever which has 3 positions as shown in above figure; Forward, Centre & Back. - PTT1 is in forward position in which PTT is ON momentary. - PTT2 is in back position in which PTT remains ON till it is taken back. - In Centre position, PTT remains OFF. - Set the PTT lever to forward position, display will show pic 3. - Set the PTT lever to back position, display will show pic 4. If PTT movement is detected & displayed correctly, PTT functionality is Ok. Press MENU to come out from BITE Test.	① Display Test Keypad Test PTT Test Mic Test ② Port:CMDHS PTT 1: Released PTT 2: Released ↑↓ ③ Port:CMDHS PTT 1: Pressed PTT 2: Released ↑↓ ④ Port:CMDHS PTT 1: Released PTT 2: Pressed ↑↓

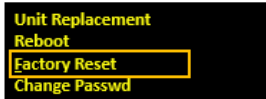
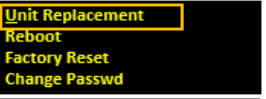
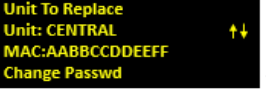
S/N	Action	Action Description	Display Content
D	MENU > Diagnostics> MIC Test	- When cursor is on Mic Test, press OK to go further or press MENU to go back. Refer picture 1. When press OK, display will show picture 2. - Press OK, display content will be as per picture 3. - Get the PTT ON by taking the PTT lever to forward position & start speaking in headset mic. - To stop recording, get the PTT lever in centre position (means PTT OFF) & press OK. Picture 4 will be displayed. - Press OK to listen the recorded voice message. - If recorded message is played back properly, MIC test is passed. - Press MENU to exit from BITE.	① Display Test Keypad Test PTT Test Mic Test ② Mic Test function Port:CMDHS OK to continue MENU Key to cancel ③ Press PTT and record OK to stop recording MENU ④ Press OK to play the recording MENU

S/N	Action	Action Description	Display Content
E	MENU > Diagnostics> BITE > Keyboard Test	- When cursor is on Keyboard Test, press Enter key on keyboard. - Display content will be as per figure 2. Press Enter again, content shown in figure 3 will be displayed. - Start pressing every key one by one. Whatever key is pressed, it will be shown in display. - For example, for character 'E' & special letter '<', display content is shown in figure 3 & 4 respectively. - For the testing of 'Fn' key, try to test any character / symbol which is printed in blue colour on keyboard. All the blue coloured characters/symbols have to be used with 'Fn' key. So if desired blue coloured character/symbol is shown in display content, 'Fn' key also is passed. - After testing all the keys, press CMD key to exit.	① Keyboard Test Enter ② Keyboard Test OK to continue MENU Key to cancel Enter ③ Press desired key Key is Alphabet - E Press E ④ Press desired key Key is Symbol - < Press <

Table 21: Diagnostics

7.2.6 Maintenance

This section is to do unit replacement, reboot, reset (hard/soft) the iVIS,

S/N	Action	Action Description	Display Content
A	MENU > Maintenance > Unit Replacement	- This option is used when any deployed user unit is supposed to be replaced with any other user unit. Another unit can be a fresh unit or pre-configured (not known) unit. - Following steps need to follow for unit replacement. - Connect new unit to the CSU using data power cable. - Give a Factory Reset to new unit (refer section C of this table). - New unit will be restarted & the content shown on its display will be as figure 2. - New unit will try to fetch the configuration from CSU. MAC Id also will be displayed on display. - Now go to any other functional user unit & follow the steps; - Press MENU > Maintenance > Unit Replacement (after entering correct password) & press OK. - Unit will start showing all the units (including CSU) one by one along with associated MAC Id. First CSU will be displayed (refer figure 4).	On new connected User Unit ①  OK ②  On functional User Unit ③  OK ④ 

S/N	Action	Action Description	Display Content
		- Use arrow key to view next unit. While viewing the desired unit which is supposed to configure, press OK. Content shown in figure 6 will be displayed. - For example, if CMDR unit is selected, figure 6 will show CMDR unit and its currently associated MAC Id. - Use arrow key, MAC ID of new user unit also will be displayed. In case 2 new user units are connected, MAC Id of both new units shall be displayed. Press OK. - Selected MAC Id will get associated with CMDR & display content will be as shown in figure 7. - Look at the new user unit. New unit will start getting the configuration settings from CSU & will start working after few seconds. ▪ New User Unit has been configured to work in place of old unit. Note: User Unit, which is used to configure the Unit Replacement, cannot replace himself. For example, if a functional unit (say GUNR) wants to replace the MAC Id of GUNR, iVIS will reject this reject as shown in figure 8.	5 Unit: To Replace Unit: CMDR MAC:AABBCCDDEEFF Change Passwd ↑↓ OK 6 Unit: CMDR MAC:AABBCCDDEEFF OK 7 Unit Replaced Press any Key to Exit OK 8 Unit: GUNR You Can't Replace Yourself OK

S/N	Action	Action Description	Display Content
B	MENU > Maintenance > Reboot	▪ Select MENU > Maintenance (refer figure 1), press OK. ▪ User will be prompted for a password as shown in figure 2. After entering correct password, 4 options shall be displayed to user as shown in figure 3. ▪ Use arrow key to select 'Reboot' & press OK. ▪ Press OK to reboot the iVIS else press MENU to exit. ▪ If pressed OK, iVIS will be restarted & configuration will remain intact.	① Diagnostics Maintenance OK ② Enter Password: **** MENU OK OK ③ Unit Replacement Reboot Factory Reset Change Passwd OK ④ Reboot the Unit? MENU OK OK Rebooting....

S/N	Action	Action Description	Display Content
C	MENU > Maintenance > Factory Reset	- Select MENU > Maintenance (refer figure 1), press OK. - User will be prompted for a password as shown in figure 2. After entering correct password, 4 options shall be displayed to user as shown in figure 3. - Use arrow key to select 'Factory Reset' & press OK. - Press OK to reset the iVIS else press MENU to exit. - If pressed OK, iVIS will get reset & iVIS configuration also will get reset to factory settings.	① Diagnostics Maintenance OK ② Enter Password: **** MENU OK OK ③ Unit Replacement Reboot Factory Reset Change Passwd OK ④ Reset the Unit? MENU OK OK ⑤ Reset Done Restarting... OK

S/N	Action	Action Description	Display Content
D	MENU > Maintenance > Change Passwd	- Select MENU > Maintenance (refer figure 1), press OK. User will be prompted for a password. After entering correct password, 4 options shall be displayed to user as shown in figure 1. - Use arrow key to select 'Change Passwd' & press OK. Content shown in figure 2 will be displayed. - Enter the desired password. Password is combination of function keys i.e. F1, F2, F3 & F4. Min length of password is 4 & max is 6. For ex, password can be, - Password with length 4 : F3 F2 F4 F1 - Password with length 5 : F1 F2 F4 F3 F2 - Password with length 6 : F1 F2 F1 F2 F4 F2 - Press OK. System will prompt to re-enter the password as shown in figure 3. - If same password is re-entered, password will be updated as shown in figure 4. Note: Default password is F1 F2 F3 F4.	① Unit Replacement Reboot Factory Reset Change Passwd OK ② Enter New Password: Range 4-6 ***** MENU OK OK ③ ReEnter Password: ***** MENU OK OK ④ PASSWORD UPDATED MENU OK

Table 22: Maintenance

8 Document History

Rev.	Change	Date	Author	Approved by
PA1	First Release	10/07/26	SG	PS

Table 23: Document Change History

RESTRICTED

PERSONAL NOTES

RESTRICTED

RESTRICTED

PERSONAL NOTES

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PERSONAL NOTES

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